Venous Thromboembolism Prophylaxis in Hospitalized Medical Patients and Those With Stroke: A Background Review for an American College of Physicians Clinical Practice Guideline

Frank A. Lederle, MD; Dylan Zylla, MD; Roderick MacDonald, MS; and Timothy J. Wilt, MD, MPH

Background: Venous thromboembolism prophylaxis has been recommended for nonsurgical patients, but its effectiveness remains uncertain.

Purpose: To assess the benefits and harms of prophylaxis in hospitalized adult medical patients and those with acute stroke.

Data Sources: MEDLINE and the Cochrane Library from 1950 through April 2011, reference lists, and study authors.

Study Selection: English-language randomized trials were included if they provided clinical outcomes and evaluated therapy with low-dose heparin or related agents or mechanical measures compared with placebo, no treatment, or other active prophylaxis in the target population.

Data Extraction: Two independent investigators extracted data on study characteristics and clinical outcomes up to 120 days after randomization. The primary outcome was total mortality.

Data Synthesis: In medical patients, heparin prophylaxis did not reduce total mortality but did result in fewer pulmonary embolisms (PEs) (odds ratio [OR], 0.69 [95% CI, 0.52 to 0.90], but with evidence of publication bias) and an increase in all bleeding events (risk ratio [RR], 1.34 [CI, 1.08 to 1.66]). Heparin prophylaxis had no statistically significant effect on any outcome in patients with acute stroke except for an increase in major bleeding events (OR, 1.66 [CI, 1.20 to 2.28]). When trials of medical patients and those with stroke were considered together (18 studies; 36 122 patients), heparin prophylaxis reduced the incidence of PE (OR, 0.70 [CI, 0.56 to 0.87]; absolute reduction, 3 events per 1000 patients treated [CI, 1 to 5 events]) but increased the incidence of all bleeding (RR, 1.28 [CI, 1.05 to 1.56]) and major bleeding events (OR, 1.61 [CI, 1.23 to 2.10]), with an absolute increase of 9 bleeding events per 1000 patients treated (CI, 2 to 18 events), 4 of which were major (CI, 1 to 7 events). A reduction in total mortality approached statistical significance (RR, 0.93 [CI, 0.86 to 1.00]; $P = 0.056$; absolute decrease, 6 deaths per 1000 patients treated [CI, 0 to 11 deaths]). No statistically significant differences in clinical outcomes were observed in the 14 trials that compared unfractionated heparin with low-molecular-weight heparin. No improvements in clinical outcomes were seen in the 3 studies of mechanical prophylaxis in patients with stroke, but more patients had lower-extremity skin damage (RR, 4.02 [CI, 2.34 to 6.91])—an increase of 39 events per 1000 patients treated (CI, 17 to 77 events).

Limitation: Non–English-language studies were not included, but these were few and small.

Conclusion: Heparin prophylaxis had no significant effect on mortality, may have reduced PE in medical patients and all patients combined, and led to more bleeding and major bleeding events, thus resulting in little or no net benefit. No differences in benefits or harms were found according to type of heparin used. Mechanical prophylaxis provided no benefit and resulted in clinically important harm to patients with stroke.

Primary Funding Source: American College of Physicians.

Ann Intern Med. 2011;155:602-615. www.annals.org
For author affiliations, see end of text.

Thirteen million nonsurgical patients are discharged each year from acute care hospitals in the United States (1). Venous thromboembolism (VTE) prophylaxis with low-dose subcutaneous anticoagulants (such as unfractionated heparin [UFH], low-molecular-weight heparin [LMWH], or fondaparinux) or mechanical leg compression has been recommended for many of these patients (2) and adopted as an optional core measure by the Joint Commission on accreditation of hospitals (3), but its effectiveness remains uncertain. Several randomized trials (4–7) have reported reductions in asymptomatic deep venous thrombosis (DVT) from heparin prophylaxis in hospitalized medical patients, but this surrogate outcome is much more common than clinically evident disease, and its value has been questioned (8–10). A quasi-randomized study (11) reported that heparin prophylaxis reduced mortality in hospitalized medical patients, but subsequent properly randomized trials have not found statistically significant reductions in mortality and have generally reported no improvement in clinically evident outcomes.

We systematically reviewed randomized trials that addressed the benefits and risks of VTE prophylaxis in hospitalized adult nonsurgical patients, including trials done in various categories of medical patients and those done in patients with acute stroke. This review serves as the back-

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ground paper for an American College of Physicians Clinical Practice Guideline.

METHODS

We developed and followed a standard protocol for all steps of the review process.

Data Sources and Selection

We searched MEDLINE from 1950 through April 2011 for articles listed in all 3 of the following Medical Subject Heading groupings: *pulmonary embolism*, *thromboembolism*, *thrombosis*, *thrombophlebitis*, *venous thromboembolism*, *venous thrombosis*, or *thromboembolism* as keywords; primary prevention, *prevent* as a keyword, or subheading "prevention and control"; and publication type (randomized, controlled trial; controlled clinical trial; phase 3 or 4 clinical trial; or meta-analysis [used to identify additional trials]). Similar terms were used to search the Cochrane Library for clinical trials of VTE prophylaxis. The titles and abstracts of identified references were reviewed, and reference lists of pertinent randomized trials and systematic reviews were used to identify additional reports. When necessary, additional information was requested from the original authors.

Study Selection

We included randomized, English-language trials that studied treatments commonly recommended and used to prevent VTE, such as subcutaneous low-dose UFH (<20 000 U/d) or similar prophylactic dosages of LMWH or related agents (such as fondaparinux) or graduated compression stockings or other mechanical measures (such as intermittent pneumatic compression). Eligible comparators included placebo, no treatment, or other active prophylaxis. Studies were limited to those that provided separate data for hospitalized medical patients or those with stroke (excluding surgical, trauma, obstetric, or pediatric patients) in acute or intermediate care facilities. Quasi-randomized trials were excluded, as were trials in which patients were routinely treated with another anticoagulant (such as warfarin or drotrecogin alfa) or thrombolytic agent. Studies of heparin (but not mechanical) prophylaxis that were done specifically in patients with acute myocardial infarction were also excluded because full-dose anticoagulation has been shown to be beneficial in acute coronary syndromes for reasons other than VTE prophylaxis; however, we did not exclude studies of medical patients that may have included some patients with the acute coronary syndrome.

Data Extraction and Quality Assessment

Two reviewers extracted the data. Discrepancies were infrequent and were resolved through discussion with all authors. Our primary outcome was total mortality up to 120 days after randomization. Secondary outcomes (all to 120 days) included symptomatic DVT; all pulmonary embolism (PE); fatal PE; all bleeding events; major bleeding events; and, for mechanical prophylaxis, skin damage. We chose 120 days to include most events likely to be related to both the intervention and any rebound effect after its discontinuation (12) without excessive dilution by unrelated later events. Because some studies reported PEs that were believed to be fatal (5), others reported all PEs detected at autopsy (13), and still others reported both categories (8), we analyzed both "fatal PE" and "all PE associated with death." "All bleeding" and "major bleeding" events were, of necessity, included as defined and reported by the original authors. Studies that gave criteria for major bleeding events usually included bleeding that was fatal, required surgery, was in a critical location or caused permanent injury, or was overt and involved transfusion of more than 2 units of blood. A recent systematic review (14) found that major bleeding events, similarly defined, had a similar risk for death to that of recurrent symptomatic VTE. For studies of stroke, we followed the practice of most of the original authors and included hemorrhagic transformations of stroke as "bleeding" and, if symptomatic, as "major bleeding." When possible, we included all events after randomization according to intention to treat, even if the original authors had excluded them.

We excluded surrogate outcomes, such as asymptomatic DVT or thrombocytopenia (a very rarely reported finding) detected by routine testing of study patients, as well as diagnoses of VTE that were based only on history and physical examination without confirmatory testing.

Study quality was assessed on the basis of the adequacy of allocation concealment (15), blinding methods (patient, investigator, or outcome assessor), intention-to-treat analysis, adequate details of study withdrawals and dropouts, and source of funding. The overall strength of evidence for the randomized trials was evaluated by using methods developed by the Agency for Healthcare Research and Quality and the Effective Health Care Program (16). The strength of the evidence was evaluated on the basis of 4 required domains: risk for bias (whether the studies for a given outcome or comparison had good internal validity), consistency (the degree of similarity in the effect sizes of the included studies), directness (whether the studies showed a single, direct link between the intervention of interest and the outcome), and precision (the degree of certainty surrounding an effect estimate of a given outcome).

Data Synthesis and Analysis

We conducted meta-analyses on comparisons that were made in multiple studies, including prophylaxis with heparin and related agents versus no heparin, prophylaxis with LMWH versus UFH, and mechanical prophylaxis versus no mechanical prophylaxis. Studies of patients with stroke were analyzed both separately from and together with studies of medical patients. Data were pooled and analyzed by using Cochrane Collaboration Review Manager software (The Nordic Cochrane Center, Copenhagen, Denmark). Heterogeneity between studies was assessed by using chi-square and I^2 tests, with a P value less than 0.100 and an I^2 value greater than 50% considered high (17).

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Because of the frequent zero rates for various outcomes in small studies, Peto odds ratios (ORs) were used except when event rates exceeded 1% (for mortality, all bleeding events, and skin damage), in which case risk ratios (RRs) from random-effects models were used (18, 19). Absolute measures of intervention effects were calculated from ORs or RRs and pooled risk in control participants by using GRADEpro, version 3.2 (http://ims.cochrane.org/gradepro).

Various sensitivity analyses were performed to assess the robustness of our findings. The analyses of heparin trials were limited to only double-blinded studies to evaluate the importance of subjectivity in the ascertainment of outcomes other than death. Analyses were also repeated after excluding heparin dosages that were no longer recommended for prophylaxis and with follow-up limited to 90 days. We examined funnel plots and performed Egger intercept tests (20) to assess for publication bias.

Role of the Funding Source

The American College of Physicians Clinical Guidelines Committee supported this project. Committee members assisted in the formulation of the questions, reviewed drafts of this report, recommended changes, and approved submission.

RESULTS

Identified Studies

Our search for randomized trials yielded 4340 references (**Appendix Figure 1**, available at www.annals.org). After exclusions based on a review of the abstracts, 302 articles were obtained for further review; 40 unique randomized trials that met our inclusion criteria, involving more than 52 000 patients, were ultimately identified. Of these, 21 were conducted in medical patients, of which 10 compared prophylaxis with heparin or related agents with no heparin (4–8, 10, 13, 21–23), 9 compared LMWH with UFH (24–32), and 2 compared different durations of LMWH prophylaxis (33, 34). Nineteen trials were conducted in patients with acute stroke, of which 8 compared heparin prophylaxis with no heparin (35–42), 5 compared LMWH with UFH (43–47), 3 compared mechanical with no mechanical prophylaxis (48–50), and 3 had unique designs (51–53). **Appendix Table 1** (available at www.annals.org) shows the characteristics of these studies.

We identified 3 pairs of studies (24, 26, 28, 29, 35, 36) for which dual publication of some patient data seemed possible on the basis of similarities in design, location, and personnel. In all 3 instances, the original authors confirmed that no dual publication occurred, and all 6 studies were included. For several studies (4, 27), we included deaths after randomization that the original authors had excluded. We excluded outcomes without confirmatory testing (3 cases of "fatal PE") from 1 study (22). One study (5) randomly assigned patients to receive placebo or 1 of 2 prophylactic doses of enoxaparin, which we combined into 1 treatment group. In 2 studies that compared patients receiving low- or high-dose heparin with control participants (41, 42), the high-dose group was excluded because it exceeded our threshold for prophylactic dosing.

Entry criteria were typically designed to enroll patients believed to be at increased risk for VTE, but none of the studies used risk assessment scales or scores. Patients in the medical trials usually had to be above a minimum age, typically 60 years, but the mean age was usually more than 70 years (**Appendix Table 1**). Studies varied on whether any patient admitted to a medical service was eligible or only patients who received certain diagnoses, and whether a minimum duration of immobilization or hospital stay was expected.

Patients were generally excluded from these trials if they had a definite indication for heparin (such as current thromboembolism or atrial fibrillation) or contraindication to it (such as high bleeding risk or thrombocytopenia). Other usual exclusions were recent surgery, recent myocardial infarction or stroke (if eligibility was not already limited to specific diagnoses), and expected short survival. Trials of patients with stroke had similar entry criteria to those of medical patients, frequently specified the duration (typically no more than 1 to 3 days) or severity of stroke, and usually excluded patients who had hemorrhagic stroke or were comatose. Studies of mechanical prophylaxis had no exclusions related to coagulation or bleeding status. The mean or median time receiving prophylaxis was 6 to 14 days in included studies, although outcomes were often collected for a longer period.

Several trials examined subgroup effects, including age, sex, medical diagnoses, and characteristics of stroke (29, 42, 47, 50, 54, 55), but none reported significant interactions or differences among any subgroups. Several medical studies enrolled only patients with 1 condition, including cancer (23), infectious disease (8), or mechanical ventilation due to exacerbation of chronic obstructive lung disease (6), so they provided indication-specific results.

Of the 18 trials that compared heparin with no heparin, 44% reported adequate allocation concealment, 61% were double-blind, 56% analyzed by intention to treat, 78% adequately reported reasons and proportions of withdrawals or dropouts, and 39% reported nonindustry funding (**Appendix Figure 2**, available at www.annals.org). Of the 14 trials of LMWH versus UFH, 36% reported adequate allocation concealment, 71% were double-blind, 29% analyzed by intention to treat, 86% adequately reported reasons and proportions of withdrawals or dropouts, and 1 reported funding by a nonindustry source.

Heparin Versus No Heparin

The **Table** and **Figure 1** show the outcomes for prophylaxis with heparin or related agents compared with no heparin. All studies used UFH or LMWH, except 1 that used fondaparinux (22), so we use the term "heparin" to describe these studies. In medical patients (10 trials; 20 717 participants), heparin prophylaxis did not significantly reduce the primary outcome of total mortality (RR,

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Table. **Outcomes of Venous Thromboembolism Prophylaxis in Nonsurgical Patients**

Group and Outcome	Studies, *n*	Intervention Group, *n/N (%)*	Control Group, *n/N (%)*	Peto Odds Ratio (95% CI)*	Absolute Effect per 1000 Patients Treated (95% CI)
		Heparin	No Heparin		
Heparin vs. no heparin					
Medical patients					
Mortality	10	679/10 466 (6.5)	679/10 251 (6.6)	0.94 (0.84 to 1.04)	−4 (−11 to 3)
Symptomatic DVT	5	25/3166 (0.79)	27/2791 (0.96)	0.78 (0.45 to 1.35)	−2 (−6 to 4)
PE	10	88/10 466 (0.84)	127/10 251 (1.2)	0.69 (0.52 to 0.90)	−4 (−6 to −1)
PE associated with death	6	50/10 157 (0.49)	53/9937 (0.53)	0.93 (0.63 to 1.38)	0 (−2 to 2)
Fatal PE	5	21/8927 (0.24)	26/8693 (0.30)	0.77 (0.43 to 1.37)	−1 (−2 to 1)
All bleeding events	8	216/4550 (4.7)	115/4194 (2.7)	1.34 (1.08 to 1.66)	9 (2 to 18)
Major bleeding events	9	41/10 331 (0.40)	25/10 116 (0.25)	1.49 (0.91 to 2.43)	1 (0 to 3)
Patients with stroke					
Mortality	8	496/5276 (9.4)	990/10 129 (9.8)	0.91 (0.70 to 1.18)	−9 (−29 to 18)
Symptomatic DVT	1	0/101	1/105 (0.95)	0.14 (0.00 to 7.09)	−9 (−10 to 57)
PE	5	39/5015 (0.78)	95/9847 (0.96)	0.72 (0.50 to 1.04)	−3 (−5 to 0)
PE associated with death	2	32/5004 (0.64)	72/9879 (0.73)	0.70 (0.46 to 1.05)	−2 (−4 to 0)
Fatal PE	2	25/4912 (0.51)	40/9769 (0.41)	1.25 (0.74 to 2.09)	1 (−1 to 4)
All bleeding events	6	24/272 (8.8)	25/250 (10)	0.95 (0.55 to 1.63)	−5 (−45 to 53)
Major bleeding events	8	79/5276 (1.5)	89/10 129 (0.88)	1.66 (1.20 to 2.28)	6 (2 to 12)
All patients combined					
Mortality	18	1175/15 742 (7.5)	1669/20 380 (8.2)	0.93 (0.86 to 1.00)	−6 (−11 to 0)
Symptomatic DVT	6	25/3267 (0.77)	28/2896 (0.97)	0.75 (0.43 to 1.30)	−2 (−6 to 3)
PE	15	127/15 481 (0.82)	222/20 098 (1.1)	0.70 (0.56 to 0.87)	−3 (−5 to −1)
PE associated with death	8	82/15 161 (0.54)	125/19 816 (0.63)	0.81 (0.61 to 1.08)	−1 (−2 to 0)
Fatal PE	7	46/13 839 (0.33)	66/18 462 (0.36)	1.01 (0.68 to 1.48)	0 (−1 to 2)
All bleeding events	14	240/4822 (5.0)	140/4444 (3.2)	1.28 (1.05 to 1.56)	9 (2 to 18)
Major bleeding events	17	120/15 607 (0.77)	114/20 245 (0.58)	1.61 (1.23 to 2.10)	4 (1 to 7)
		LMWH	UFH		
LMWH vs. UFH					
Medical patients					
Mortality	9	539/5816 (9.3)	594/5834 (10.2)	0.91 (0.73 to 1.13)	−9 (−28 to 13)
Symptomatic DVT	3	3/1878 (0.16)	3/1875 (0.16)	1.00 (0.20 to 4.94)	0 (−2 to 8)
PE	9	30/5816 (0.52)	43/5834 (0.74)	0.70 (0.44 to 1.11)	−2 (−4 to 1)
PE associated with death	2	1/1072 (0.09)	0/1068	7.40 (0.15 to 373.14)	0 (0 to 0)
Fatal PE	3	4/2698 (0.15)	3/2686 (0.11)	1.33 (0.30 to 5.84)	0 (−1 to 5)
All bleeding events	8	328/5767 (5.9)	365/5784 (6.5)	0.90 (0.76 to 1.05)	−7 (−16 to 3)
Major bleeding events	9	120/5816 (2.1)	134/5834 (2.3)	0.89 (0.70 to 1.15)	−3 (−7 to 3)
Patients with stroke					
Mortality	5	168/1396 (12.0)	166/1389 (12.0)	1.00 (0.81 to 1.22)	0 (−23 to 26)
Symptomatic DVT	2	3/990 (0.30)	10/984 (1.0)	0.34 (0.11 to 1.00)	−7 (−9 to 0)
PE	4	8/1351 (0.59)	14/1347 (1.0)	0.57 (0.25 to 1.34)	−4 (−8 to 3)
PE associated with death	4	6/1351 (0.44)	6/1347 (0.45)	1.00 (0.32 to 3.13)	0 (−3 to 9)
Fatal PE	4	6/1351 (0.44)	6/1347 (0.45)	1.00 (0.32 to 3.13)	0 (−3 to 9)
All bleeding events	5	87/1396 (6.2)	91/1389 (6.6)	0.95 (0.70 to 1.28)	−3 (−20 to 18)
Major bleeding events	5	18/1396 (1.3)	12/1389 (0.86)	1.49 (0.73 to 3.06)	4 (−2 to 19)
All patients combined					
Mortality	14	707/7212 (9.8)	760/7223 (10.5)	0.94 (0.82 to 1.08)	−6 (−19 to 8)
Symptomatic DVT	5	6/2868 (0.21)	13/2859 (0.45)	0.47 (0.19 to 1.17)	−3 (−4 to 1)
PE	13	38/7167 (0.53)	57/7181 (0.79)	0.67 (0.45 to 1.00)	−3 (−4 to 0)
PE associated with death	6	7/2423 (0.29)	6/2415 (0.25)	1.17 (0.39 to 3.50)	0 (−1 to 5)
Fatal PE	7	10/4049 (0.25)	9/4033 (0.22)	1.11 (0.45 to 2.74)	0 (−1 to 3)
All bleeding events	13	415/7163 (5.9)	456/6999 (6.5)	0.92 (0.81 to 1.04)	−5 (−12 to 3)
Major bleeding events	14	138/7212 (1.9)	146/7223 (2.0)	0.95 (0.75 to 1.20)	−1 (−5 to 4)
		Mechanical Prophylaxis	No Mechanical Prophylaxis		
Mechanical vs. no mechanical prophylaxis					
Mortality	3	133/1334 (10.0)	114/1307 (8.7)	1.13 (0.89 to 1.43)	11 (−10 to 37)
Symptomatic DVT	1	55/1256 (4.4)	61/1262 (4.8)	0.91 (0.63 to 1.29)	−4 (−18 to 14)
PE	2	13/1321 (0.98)	20/1294 (1.5)	0.65 (0.33 to 1.31)	−5 (−10 to 5)
Skin damage (ulcers, blisters, necrosis)	1	64/1256 (5.1)	16/1262 (1.3)	4.02 (2.34 to 6.91)	39 (17 to 77)

DVT = deep venous thrombosis; LMWH = low-molecular-weight heparin; PE = pulmonary embolism; UFH = unfractionated heparin.
* Risk ratio was used when event rates were >1% (mortality, any bleeding events, and skin damage).

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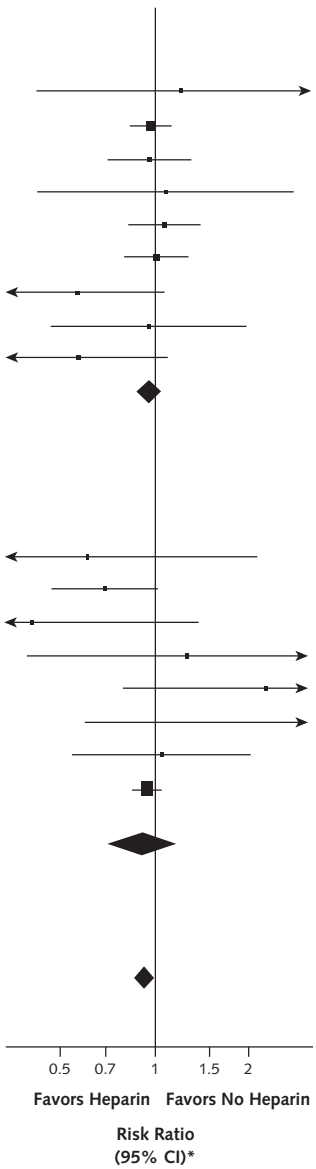
Figure 1. **Forest plots for outcomes of death, pulmonary embolism, and major bleeding events in studies of heparin prophylaxis versus no heparin.**

Outcome and Study, Year (Reference)	Heparin Events, *n*	Heparin Patients, *n*	No Heparin Events, *n*	No Heparin Patients, *n*	Weight, %	Risk Ratio (95% CI)*
Death						
Medical patients						
Belch et al, 1981 (21)	0	50	0	50		Not estimable
Dahan et al, 1986 (4)	7	135	6	135	0.5	1.17 (0.40–3.38)
Gärdlund, 1996 (8)	304	5776	333	5917	23.2	0.94 (0.80–1.09)
Samama et al, 1999 (5)	97	731	53	371	5.9	0.93 (0.68–1.27)
Fraisse et al, 2000 (6)	8	109	8	114	0.7	1.05 (0.41–2.69)
Leizorovicz et al, 2004 (7)	107	1856	103	1850	8.2	1.04 (0.80–1.35)
Mahé et al, 2005 (13)	124	1230	128	1244	10.3	0.98 (0.78–1.24)
Cohen et al, 2006 (22)	14	429	25	420	1.4	0.55 (0.29–1.04)
Lederle et al, 2006 (10)	13	140	14	140	1.1	0.93 (0.45–1.90)
Weber et al, 2008 (23)	5	10	9	10	1.4	0.56 (0.29–1.07)
Subtotal		10 466		10 251	52.7	0.94 (0.84–1.04)
Total events	679		679			
Heterogeneity: $\tau^2 = 0.00$, chi-square = 6.08 ($P = 0.64$), $I^2 = 0\%$						
Test for overall effect: $Z = 1.28$ ($P = 0.20$)						
Patients with stroke						
McCarthy et al, 1977 (35)	3	16	5	16	0.4	0.60 (0.17–2.10)
McCarthy and Turner, 1986 (36)	31	144	51	161	3.9	0.68 (0.46–1.00)
Turpie et al, 1987 (37)	4	50	5	25	0.4	0.40 (0.12–1.36)
Dickmann et al, 1988 (38)	5	23	4	23	0.4	1.25 (0.38–4.07)
Prins et al, 1989 (39)	9	30	4	30	0.5	2.25 (0.78–6.52)
Sandset et al, 1990 (40)	5	52	1	51	0.1	4.90 (0.59–40.53)
Kay et al, 1995 (41)	15	101	15	105	1.3	1.04 (0.54–2.01)
International Stroke Trial Collaborative Group, 1997 (42)	424	4860	905	9718	40.3	0.94 (0.84–1.05)
Subtotal		5276		10 129	47.3	0.91 (0.70–1.18)
Total events	496		990			
Heterogeneity: $\tau^2 = 0.04$, chi-square = 10.23 ($P = 0.18$), $I^2 = 32\%$						
Test for overall effect: $Z = 0.72$ ($P = 0.47$)						
Total		15 742		20 380	100.0	0.93 (0.86–1.00)
Total events	1175		1669			
Heterogeneity: $\tau^2 = 0.00$, chi-square = 16.34 ($P = 0.43$), $I^2 = 2\%$						
Test for overall effect: $Z = 1.91$ ($P = 0.060$)						
Test for subgroup differences: chi-square = 0.04 ($P = 0.83$), $I^2 = 0\%$						



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0.94 [CI, 0.84 to 1.04]) but did result in fewer PEs (OR, 0.69 [CI, 0.52 to 0.90]; absolute reduction, 4 events per 1000 patients treated [CI, 1 to 6 events]) and more bleeding events (RR, 1.34 [CI, 1.08 to 1.66]; absolute increase, 9 events per 1000 patients treated [CI, 2 to 18 events]). Heparin prophylaxis did not significantly affect any outcomes in patients with acute stroke (8 trials; 15 405 participants), except for an increase in major bleeding events (OR, 1.66 [CI, 1.20 to 2.28]; absolute increase, 6 events per 1000 patients treated [CI, 2 to 12 events]).

When all trials of heparin prophylaxis versus no heparin in nonsurgical patients were combined (18 trials; 36 122 participants), the reduction in mortality approached but did not reach statistical significance (RR, 0.93 [CI, 0.86 to 1.00]; $P = 0.056$). Heparin therapy decreased the incidence of PE (OR, 0.70 [CI, 0.56 to 0.87]; absolute reduction, 3 events per 1000

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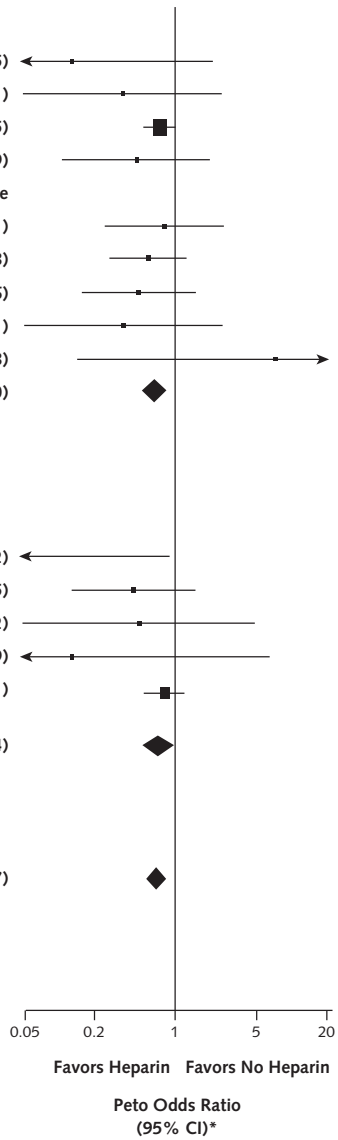
Figure 1—**Continued**

Outcome and Study, Year (Reference)	Heparin Events, *n*	Heparin Patients, *n*	No Heparin Events, *n*	No Heparin Patients, *n*	Weight, %	Peto Odds Ratio (95% CI)*
Pulmonary embolism						
Medical patients						
Belch et al, 1981 (21)	0	50	2	50	0.6	0.13 (0.01–2.15)
Dahan et al, 1986 (4)	1	135	3	135	1.2	0.36 (0.05–2.61)
Gärdlund, 1996 (8)	62	5776	84	5917	44.0	0.76 (0.54–1.05)
Samama et al, 1999 (5)	4	731	4	371	2.2	0.48 (0.11–2.09)
Fraisse et al, 2000 (6)	0	109	0	114		Not estimable
Leizorovicz et al, 2004 (7)	5	1856	6	1850	3.3	0.83 (0.25–2.71)
Mahé et al, 2005 (13)	10	1230	17	1244	8.1	0.60 (0.28–1.28)
Cohen et al, 2006 (22)	4	429	8	420	3.6	0.50 (0.16–1.56)
Lederle et al, 2006 (10)	1	140	3	140	1.2	0.36 (0.05–2.61)
Weber et al, 2008 (23)	1	10	0	10	0.3	7.39 (0.15–372.38)
Subtotal		10 466		10 251	64.6	0.69 (0.52–0.90)
Total events	88		127			
Heterogeneity: chi-square = 4.63 (P = 0.80), I^2 = 0%						
Test for overall effect: Z = 2.74 (P = 0.006)						
Patients with stroke						
Turpie et al, 1987 (37)	0	50	2	25	0.5	0.05 (0.00–0.92)
Dickmann et al, 1988 (38)	5	23	9	23	3.0	0.45 (0.13–1.55)
Prins et al, 1989 (39)	1	30	2	30	0.9	0.50 (0.05–5.02)
Sandset et al, 1990 (40)	0	52	1	51	0.3	0.13 (0.00–6.69)
International Stroke Trial Collaborative Group, 1997 (42)	33	4860	81	9718	30.7	0.82 (0.55–1.21)
Subtotal		5015		9847	35.4	0.72 (0.50–1.04)
Total events	39		95			
Heterogeneity: chi-square = 5.02 (P = 0.29), I^2 = 20%						
Test for overall effect: Z = 1.73 (P = 0.080)						
Total		15 481		20 098	100.0	0.70 (0.56–0.87)
Total events	127		222			
Heterogeneity: chi-square = 9.71 (P = 0.72), I^2 = 0%						
Test for overall effect: Z = 3.23 (P = 0.001)						
Test for subgroup differences: chi-square = 0.06 (P = 0.81), I^2 = 0%						



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patients treated [CI, 1 to 5 events]). Heparin prophylaxis increased all bleeding events (RR, 1.28 [CI, 1.05 to 1.56]) and major bleeding events (OR, 1.61 [CI, 1.23 to 2.10]); a statistically significant, absolute increase of 9 bleeding events per 1000 patients treated (CI, 2 to 18 events), 4 of them major (CI, 1 to 7 events), occured.

Other Comparisons

The **Table** and **Figure 2** show event rates from randomized trials that compared LMWH with UFH (14 trials; 14 435 participants). The 2 treatments did not significantly differ for any outcome in medical studies, studies of stroke, or all studies combined. Mortality was very similar for the 2 treatments, whereas a nonsignificant difference favored LMWH for PE (OR for all patients, 0.67 [CI, 0.45 to 1.00]; P = 0.053). A decrease in symptomatic DVT with LMWH in studies of stroke approached statistical significance (P = 0.052), but only 13 events occurred.

Data on clinical outcomes from randomized trials of mechanical prophylaxis compared with no prophylaxis in

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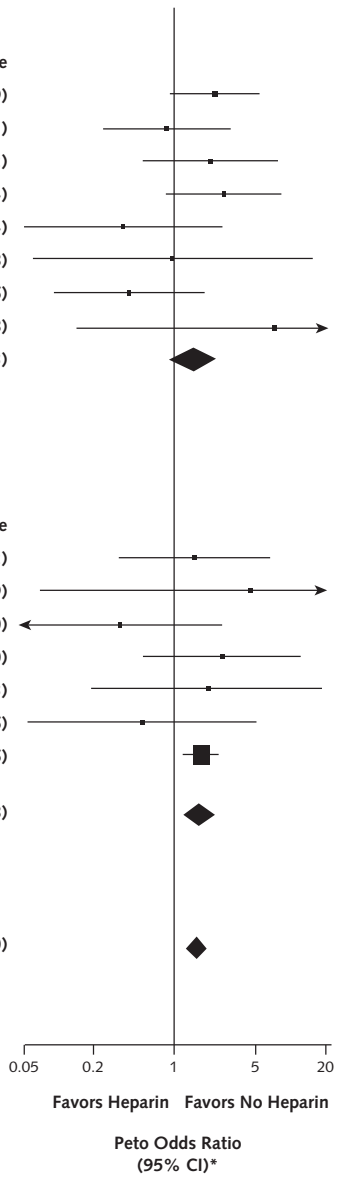
Figure 1—**Continued**

Outcome and Study, Year (Reference)	Heparin Events, *n*	Heparin Patients, *n*	No Heparin Events, *n*	No Heparin Patients, *n*	Weight, %	Peto Odds Ratio (95% CI)*
Major bleeding events						
Medical patients						
Belch et al, 1981 (21)	0	50	0	50		Not estimable
Gärdlund, 1996 (8)	14	5776	6	5917	9.3	2.28 (0.95–5.49)
Samama et al, 1999 (5)	7	731	4	371	4.6	0.89 (0.25–3.11)
Fraisse et al, 2000 (6)	6	109	3	114	4.1	2.09 (0.55–7.92)
Leizorovicz et al, 2004 (7)	9	1856	3	1850	5.6	2.72 (0.87–8.44)
Mahé et al, 2005 (13)	1	1230	3	1244	1.9	0.37 (0.05–2.64)
Cohen et al, 2006 (22)	1	429	1	420	0.9	0.98 (0.06–15.68)
Lederle et al, 2006 (10)	2	140	5	140	3.2	0.42 (0.09–1.86)
Weber et al, 2008 (23)	1	10	0	10	0.5	7.39 (0.15–372.38)
Subtotal		10 331		10 116	30.0	1.49 (0.91–2.43)
Total events	41		25			
Heterogeneity: chi-square = 8.34 (P = 0.30), I^2 = 16%						
Test for overall effect: Z = 1.60 (P = 0.110)						
Patients with stroke						
McCarthy et al, 1977 (35)	0	16	0	16		Not estimable
McCarthy and Turner, 1986 (36)	4	144	3	161	3.2	1.50 (0.34–6.72)
Turpie et al, 1987 (37)	1	50	0	25	0.4	4.48 (0.07–286.49)
Dickmann et al, 1988 (38)	1	23	3	23	1.7	0.34 (0.05–2.60)
Prins et al, 1989 (39)	5	30	2	30	2.9	2.60 (0.54–12.40)
Sandset et al, 1990 (40)	2	52	1	51	1.4	1.94 (0.20–19.03)
Kay et al, 1995 (41)	1	101	2	105	1.4	0.53 (0.05–5.16)
International Stroke Trial Collaborative Group, 1997 (42)	65	4860	78	9718	58.9	1.73 (1.22–2.46)
Subtotal		5276		10 129	70.0	1.66 (1.20–2.28)
Total events	79		89			
Heterogeneity: chi-square = 3.92 (P = 0.69), I^2 = 0%						
Test for overall effect: Z = 3.09 (P = 0.002)						
Total		15 607		20 245	100.0	1.61 (1.23–2.10)
Total events	120		114			
Heterogeneity: chi-square = 12.38 (P = 0.58), I^2 = 0%						
Test for overall effect: Z = 3.46 (P < 0.001)						
Test for subgroup differences: chi-square = 0.12 (P = 0.73), I^2 = 0%						



* Mantel–Haenszel risk ratio (random) and Peto odds ratio (fixed).

nonsurgical patients are scarce—nearly all are derived from 1 study (50) (**Appendix Table 1**). Our analysis therefore closely resembles the results of that study, which randomly assigned 2518 patients with acute stroke to wear thigh-length graduated compression stockings or no stockings. No statistically significant differences were observed in our analysis for mortality, symptomatic DVT, or PE, but mechanical prophylaxis caused more instances of lower-extremity skin damage (RR, 4.02 [CI, 2.34 to 6.91])—an absolute increase of 39 events per 1000 patients treated (CI, 17 to 77 events) (**Table** and **Figure 3**).

Sensitivity Analyses and Strength of Evidence

A funnel plot of all trials of heparin versus no heparin in all nonsurgical patients showed asymmetry (Egger intercept 2-sided P = 0.018) consistent with a publication bias against small studies that showed no reduction in the incidence of PE. No asymmetry was seen for trials that compared LMWH with UFH. Sta-

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tistical heterogeneity among studies was not high for any outcome except "PE associated with death" in studies of heparin versus no heparin in patients with stroke, a comparison of only 2 studies. Limiting analyses of heparin trials to double-blind studies yielded no substantive changes in the results. Limiting follow-up duration from 120 to 90 days changed the data from only 1 study (5) and yielded no substantial changes in our findings. Excluding dosages no longer recommended for prophylaxis removed only the group receiving enoxaparin, 20 mg, in Samama and colleagues' study (5) and yielded no substantial changes in our findings.

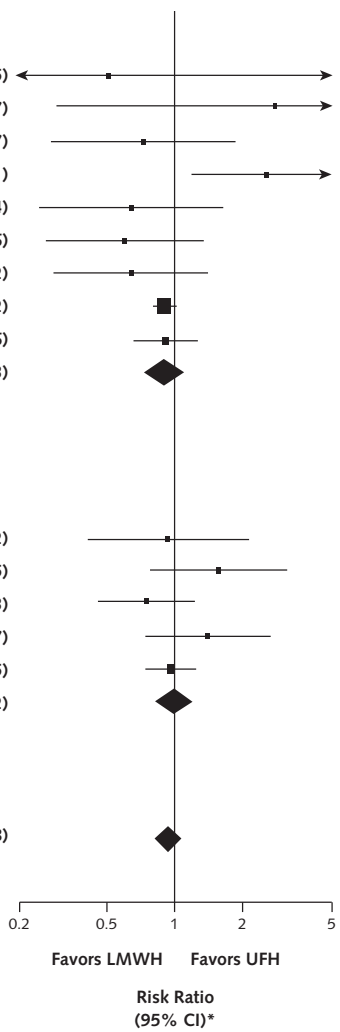
Appendix Table 2 (available at www.annals.org) summarizes our findings, including our assessment of the strength of evidence.

Miscellaneous Studies

Five studies met our inclusion criteria but were sufficiently different in their design that they could not be meaningfully combined with the other trials and so are discussed separately. Three were relatively small and found no statistically significant differences in any clinical outcomes. Luba and colleagues (33) randomly assigned 300 medical patients to receive LMWH for different durations, and they reported 2 deaths in the shorter-duration group versus none in the

Figure 2. **Forest plots for outcomes of death, pulmonary embolism, and major bleeding events in studies of low-molecular-weight versus unfractionated heparin prophylaxis.**

Outcome and Study, Year (Reference)	LMWH Events, *n*	LMWH Patients, *n*	UFH Events, *n*	UFH Patients, *n*	Weight, %	Risk Ratio (95% CI)*
Death						
Medical patients						
Aquino et al, 1990 (25)	1	49	2	50	0.3	0.51 (0.05–5.45)
Harenberg et al, 1990 (24)	3	89	1	83	0.4	2.80 (0.30–26.37)
Bergmann et al, 1996 (27)	7	217	10	225	2.0	0.73 (0.28–1.87)
Harenberg et al, 1996 (26)	23	983	9	985	3.1	2.56 (1.19–5.51)
Lechler et al, 1996 (28)	7	477	11	482	2.1	0.64 (0.25–1.64)
Kleber et al, 2003 (29)	9	332	15	333	2.7	0.60 (0.27–1.36)
Schellong et al, 2010 (31)	9	163	15	174	2.8	0.64 (0.29–1.42)
Cook et al, 2011 (30)	414	1880	459	1884	37.6	0.90 (0.80–1.02)
Riess et al, 2010 (32)	66	1626	72	1618	13.3	0.91 (0.66–1.26)
Subtotal		5816		5834	64.3	0.91 (0.73–1.13)
Total events	539		594			
Heterogeneity: $\tau^2 = 0.02$, chi-square = 10.71 ($P = 0.22$), $I^2 = 25\%$						
Test for overall effect: $Z = 0.86$ ($P = 0.39$)						
Patients with stroke						
Turpie et al, 1992 (44)	9	45	9	42	2.7	0.93 (0.41–2.12)
Dumas et al, 1994 (43)	17	89	11	90	3.6	1.56 (0.78–3.15)
Hillbom et al, 2002 (45)	21	106	28	106	6.7	0.75 (0.46–1.23)
Diener et al, 2006 (46)	21	272	15	273	4.3	1.41 (0.74–2.67)
Sherman et al, 2007 (47)	100	884	103	878	18.3	0.96 (0.74–1.25)
Subtotal		1396		1389	35.7	1.00 (0.81–1.22)
Total events	168		166			
Heterogeneity: $\tau^2 = 0.00$, chi-square = 4.04 ($P = 0.40$), $I^2 = 1\%$						
Test for overall effect: $Z = 0.02$ ($P = 0.99$)						
Total		7212		7223	100.0	0.94 (0.82–1.08)
Total events	707		760			
Heterogeneity: $\tau^2 = 0.01$, chi-square = 15.45 ($P = 0.28$), $I^2 = 16\%$						
Test for overall effect: $Z = 0.84$ ($P = 0.40$)						
Test for subgroup differences: chi-square = 0.38 ($P = 0.54$), $I^2 = 0\%$						



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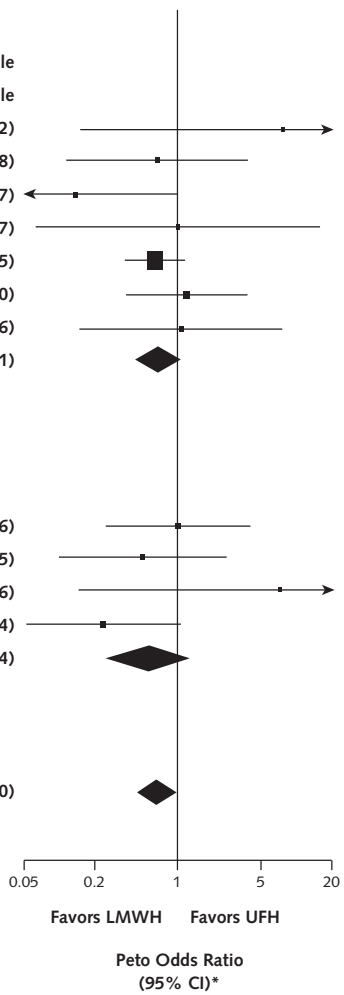
longer-duration group. Green and colleagues (51) randomly assigned 90 patients who were enrolled at least 7 days after their stroke to receive LMWH versus intermittent pneumatic compression. Most patients had received heparin prophylaxis before randomization. Two symptomatic DVTs and 1 minor bleeding event occurred in the LMWH group, compared with no events in the compression group. Lacut and colleagues (52) randomly assigned 151 patients with acute cerebral hemorrhage to wear graduated compression stockings with or without intermittent pneumatic compression, and they observed 24 deaths in the stockings-alone group and 15 in the combined treatment group, with 1 symptomatic DVT in each group. The fourth study, which involved 3114 patients with stroke randomly assigned to wear thigh-length or below-knee stockings, found no differences in clinical outcomes except for skin damage, which was less frequent with below-knee stockings (53).

In the fifth study, 6085 hospitalized, immobile medical patients who had completed an initial 10-day course of open-label enoxaparin prophylaxis were randomly assigned to receive 28 more days of enoxaparin, 40 mg, or placebo (34). After 90 days, the number of patients with symptomatic VTE in the extended prophylaxis group decreased significantly from 28 to 8, which coincided with significant increases in the number of patients with any bleeding events (assessed to 30 days) (from 116 to 186) or with major bleeding events (from 10 to 25). Total mortality at 90 days was unchanged (hazard ratio, 1.04), and the 33 episodes of VTE or major bleeding events that were considered to be serious adverse events were evenly dis-

Figure 2—**Continued**

Outcome and Study, Year (Reference)	LMWH Events, *n*	LMWH Patients, *n*	UFH Events, *n*	UFH Patients, *n*	Weight, %	Peto Odds Ratio (95% CI)*
Pulmonary embolism						
Medical patients						
Aquino et al, 1990 (25)	0	49	0	50		Not estimable
Harenberg et al, 1990 (24)	0	89	0	83		Not estimable
Bergmann et al, 1996 (27)	1	217	0	225	1.1	7.67 (0.15–386.62)
Harenberg et al, 1996 (26)	2	983	3	985	5.3	0.67 (0.12–3.88)
Lechler et al, 1996 (28)	0	477	4	482	4.2	0.14 (0.02–0.97)
Kleber et al, 2003 (29)	1	332	1	333	2.1	1.00 (0.06–16.07)
Cook et al, 2011 (30)	18	1880	28	1884	48.4	0.65 (0.36–1.15)
Riess et al, 2010 (32)	6	1626	5	1618	11.7	1.19 (0.37–3.90)
Schellong et al, 2010 (31)	2	163	2	174	4.2	1.07 (0.15–7.66)
Subtotal		5816		5834	77.1	0.70 (0.44–1.11)
Total events	30		43			
Heterogeneity: chi-square = 5.21 (P = 0.52), I^2 = 0%						
Test for overall effect: Z = 1.50 (P = 0.130)						
Patients with stroke						
Dumas et al, 1994 (43)	4	89	4	90	8.2	1.01 (0.25–4.16)
Hillbom et al, 2002 (45)	2	106	4	106	6.2	0.51 (0.10–2.55)
Diener et al, 2006 (46)	1	272	0	273	1.1	7.42 (0.15–373.76)
Sherman et al, 2007 (47)	1	884	6	878	7.4	0.24 (0.05–1.04)
Subtotal		1351		1347	22.9	0.57 (0.25–1.34)
Total events	8		14			
Heterogeneity: chi-square = 3.64 (P = 0.30), I^2 = 18%						
Test for overall effect: Z = 1.29 (P = 0.20)						
Total		7167		7181	100.0	0.67 (0.45–1.00)
Total events	38		57			
Heterogeneity: chi-square = 9.02 (P = 0.53), I^2 = 0%						
Test for overall effect: Z = 1.94 (P = 0.050)						
Test for subgroup differences: chi-square = 0.17 (P = 0.68), I^2 = 0%						



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tributed between the groups. In our assessment, the 2 durations resulted in similar clinical outcomes.

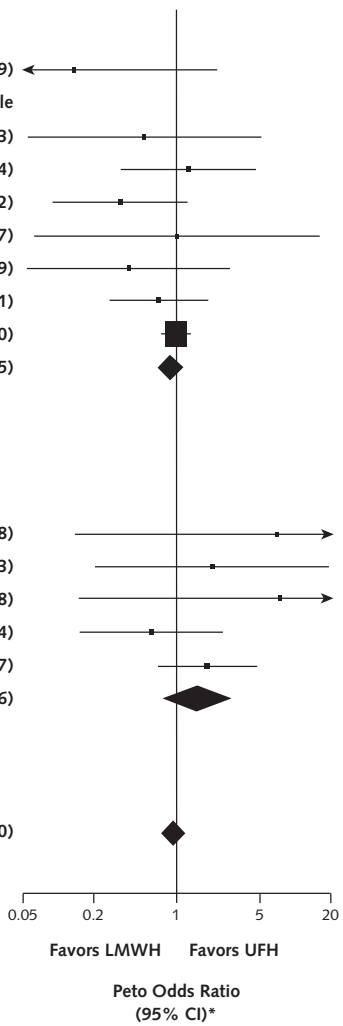
Discussion

In our systematic review of hospitalized nonsurgical patients, we found no significant effect of prophylaxis on our primary outcome of total mortality. Among secondary outcomes, we found that compared with no heparin, low-dose heparin prophylaxis prevented 3 PEs and caused 9 bleeding episodes, 4 of them major, per 1000 patients treated. The funnel plot and Egger analyses suggested that the decrease in PE incidence may have been exaggerated by publication bias. Clinical outcomes did not significantly differ between LMWH and UFH, but few symptomatic DVTs and PEs occurred, and we could not exclude potentially important differences. No improvements in clinical outcomes were seen with mechanical prophylaxis in patients with stroke, but more instances of lower-extremity skin damage occurred.

Our intent was to evaluate VTE prophylaxis in a broad range of hospitalized nonsurgical patients. Whether medical patients and those with acute stroke are similar enough to be considered together could be debated. Similar questions were addressed, the observed results were generally similar, combining studies did not add statistical heterogeneity, and the additional statistical power from combining groups was informative. However, others might prefer to consider these 2 groups

Figure 2—**Continued**

Outcome and Study, Year (Reference)	LMWH Events, *n*	LMWH Patients, *n*	UFH Events, *n*	UFH Patients, *n*	Weight, %	Peto Odds Ratio* (95% CI)
Major bleeding events						
Medical patients						
Aquino et al, 1990 (25)	0	49	2	50	0.7	0.14 (0.01–2.19)
Harenberg et al, 1990 (24)	0	89	0	83		Not estimable
Bergmann et al, 1996 (27)	1	217	2	225	1.1	0.53 (0.05–5.13)
Harenberg et al, 1996 (26)	5	983	4	985	3.3	1.25 (0.34–4.64)
Lechler et al, 1996 (28)	2	477	7	482	3.3	0.33 (0.09–1.22)
Kleber et al, 2003 (29)	1	332	1	333	0.7	1.00 (0.06–16.07)
Schellong et al, 2010 (31)	1	163	3	174	1.5	0.39 (0.05–2.79)
Riess et al, 2010 (32)	7	1626	10	1618	6.2	0.70 (0.27–1.81)
Cook et al, 2011 (30)	103	1880	105	1884	72.3	0.98 (0.74–1.30)
Subtotal		5816		5834	89.1	0.89 (0.70–1.15)
Total events	120		134			
Heterogeneity: chi-square = 5.83 (P = 0.56), I^2 = 0%						
Test for overall effect: Z = 0.87 (P = 0.38)						
Patients with stroke						
Turpie et al, 1992 (44)	1	45	0	42	0.4	6.91 (0.14–349.18)
Dumas et al, 1994 (43)	2	89	1	90	1.1	1.98 (0.20–19.33)
Hillbom et al, 2002 (45)	1	106	0	106	0.4	7.39 (0.15–372.38)
Diener et al, 2006 (46)	3	272	5	273	2.9	0.60 (0.15–2.44)
Sherman et al, 2007 (47)	11	884	6	878	6.2	1.80 (0.69–4.67)
Subtotal		1396		1389	10.9	1.49 (0.73–3.06)
Total events	18		12			
Heterogeneity: chi-square = 3.04 (P = 0.55), I^2 = 0%						
Test for overall effect: Z = 1.09 (P = 0.28)						
Total		7212		7223	100.0	0.95 (0.75–1.20)
Total events	138		146			
Heterogeneity: chi-square = 10.61 (P = 0.56), I^2 = 0%						
Test for overall effect: Z = 0.46 (P = 0.64)						
Test for subgroup differences: chi-square = 1.73 (P = 0.19), I^2 = 42.3%						



LMWH = low-molecular-weight heparin; UFH = unfractionated heparin.
* Mantel–Haenszel risk ratio (random) and Peto odds ratio (fixed).

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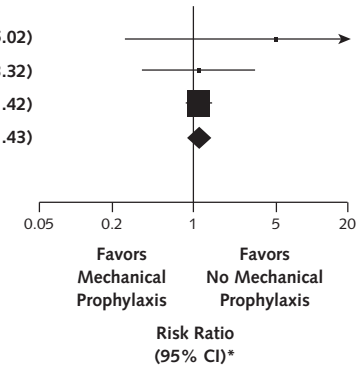
Figure 3. **Forest plot for outcome of death in studies of mechanical versus no mechanical prophylaxis.**

Outcome and Study, Year (Reference)	Mechanical Prophylaxis		No Mechanical Prophylaxis		Weight, %	Risk Ratio (95% CI)*
	Events, *n*	Patients, *n*	Events, *n*	Patients, *n*		
Death						
Prasad et al, 1982 (48)	2	13	0	13	0.7	5.00 (0.26–95.02)
Muir et al, 2000 (49)	9	65	4	32	4.7	1.11 (0.37–3.32)
Dennis et al, 2009 (50)	122	1256	110	1262	94.6	1.11 (0.87–1.42)
Total		1334		1307	100.0	1.13 (0.89–1.43)
Total events	133		114			

Heterogeneity: $\tau^2 = 0.00$, chi-square = 0.99 ($P = 0.61$), $I^2 = 0\%$

Test for overall effect: $Z = 0.97$ ($P = 0.33$)



* Mantel–Haenszel risk ratio (fixed).

separately, and our findings are formatted to enable either approach.

Previous systematic reviews have differed from ours and each other in various ways, such as by excluding studies of stroke and acute myocardial infarction (56, 57), including only studies of stroke (58), and combining studies of medical patients and those with stroke or acute myocardial infarction (59). Some (59, 60) have focused on asymptomatic DVT, which was the primary outcome of many of the individual trials. A systematic review of prevention of symptomatic VTE in hospitalized medical patients (57) differed from ours by considering only events that occurred during prophylaxis and excluding those that occurred afterward. The findings of these reviews have varied with the studies included, although the conclusions have tended to more strongly favor prophylaxis when the focus was on asymptomatic DVT. One recent review (61) compared prophylaxis with 5000 U of UFH given 2 versus 3 times daily, and it found more benefit but also more bleeding with the 3-times-daily dosage. Because we excluded asymptomatic DVT and studies of acute myocardial infarction, our data were insufficient to compare these 2 dosages.

Which outcomes should be preferred when judging the effectiveness of VTE prophylaxis remains a difficult question. The problems with surrogate outcomes in general (62), and with asymptomatic DVT in particular (8–10), have been emphasized elsewhere. Symptomatic DVT is also problematic because it is uncommon and in many cases does not precede PE. The diagnosis of PE is increasingly made by computed tomography angiography, including many small PEs that are found incidentally in 2% to 5% of hospitalized patients and are of unknown importance (63). However, the studies that we included did not screen patients with computed tomography, so the findings presented herein should reflect clinical disease. Fatal PE has been used as an outcome in some studies (8, 57), but the role of PE in a patient's death is often difficult to ascertain (63). The comparison of outcomes has been clarified recently by a systematic review finding that risk for death is similar between major bleeding events and symptomatic VTE (14). Long-term sequelae of VTE, such as pulmonary hypertension or the postthrombotic syndrome, are clinically important but impractical for use in trials because differences in these outcomes are extremely difficult to detect. Fewer than 4% of survivors of PE develop pulmonary hypertension within 2 years (64), and the clinical manifestations that define the postthrombotic syndrome occur only 1.5 times as often in patients who have had asymptomatic DVT as in control participants (65). Total mortality is clinically important and can be accurately ascertained but requires a large number of patients to demonstrate a difference because of dilution by deaths not related to VTE. Overcoming this problem is a principal reason for conducting a meta-analysis, but a statistically significant reduction in mortality from heparin prophylaxis could not be demonstrated even after all trials of hospitalized medical patients and those with stroke were combined (>36 000 patients). A very large randomized trial would be needed to properly evaluate a difference in total mortality, and 1 such trial (10) ended in the pilot phase because only a small proportion of hospitalized medical patients were candidates for enrollment. Current hospitalizations are often much shorter than the 6 to 14 days of prophylaxis we observed, which would probably reduce the risk for VTE and any benefit from prophylaxis and further increase the size of a definitive trial.

We excluded some studies that other reviews might have included, a few of which deserve explanation. Studies by Halkin and colleagues (11), Kutcher and colleagues (66), and Orken and colleagues (67) were excluded because they were quasi-randomized by medical

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record number or order of admission. Berge and colleagues' study (68) was excluded because roughly one half of the patients also received warfarin. Studies by Gallus and colleagues (69) and Pambianco and colleagues (70) were excluded because no clinical outcomes were reported in nonsurgical patients. Kierkegaard and Norgren's study (71) was excluded because randomization to mechanical prophylaxis was by leg rather than by patient and no symptomatic DVTs, the only potentially usable clinical outcome, occurred. Ibarra-Pérez and colleagues' study (72) was excluded because randomization was to 4 active treatment groups, but deaths were not reported by treatment group. Gelmers' study (73) was excluded because numbers of deaths were not given and the validity of the data was questionable (74). Finally, a large study by Levi and colleagues was excluded because all patients also received drotrecogin alfa (75).

Approximately one half of the patients in Levi and colleagues' study were receiving heparin prophylaxis before randomization, and the investigators observed that patients who were receiving heparin and were then randomly assigned to receive placebo (discontinuation of heparin therapy) had increased mortality. This is concerning in light of Gärdlund's findings (8) that patients randomly assigned to receive heparin prophylaxis for up to 21 days (mean, 8.2 days) had only one fourth as many fatal PEs as control patients at 20 days but the same number by 40 days, again consistent with a deleterious effect of stopping heparin therapy. These data suggest that the rebound thrombotic events reported after stopping full-dose heparin therapy (12) may also occur after stopping low-dose heparin therapy.

On the basis of our findings, if heparin prophylaxis is to be used in hospitalized nonsurgical patients, the choice of agent will probably depend on such factors as convenience and cost rather than comparative effectiveness and harms. All of the prophylactic anticoagulants examined in our review are given as subcutaneous injections, 2 or 3 times daily for UFH and once daily for LMWH or fondaparinux. Average wholesale drug costs are about $10 per day for UFH, $35 per day for LMWH, and $60 per day for fondaparinux. Although few data on effect size from randomized trials are available to refine patient selection, several trial reports (34, 54, 55) have suggested that heparin is less beneficial in patients younger than 75 years.

We conclude that in hospitalized medical patients, low-dose heparin prophylaxis reduced PE but not total mortality or other clinical outcomes and resulted in more episodes of bleeding. In patients with acute stroke, we found no statistically significant benefit from heparin prophylaxis but found increased risk for major bleeding. When considering medical patients and those with stroke together, low-dose heparin prophylaxis may have reduced PE and increased risk for bleeding and major bleeding events and had no statistically significant effect on mortality. We interpret these findings as indicative of little or no net benefit. No significant differences in clinical benefits or harms were found between UFH and LMWH. Mechanical prophylaxis was not effective at preventing VTE and resulted in clinically important harms.

From the Minnesota Evidence-based Practice Center and Minneapolis Veterans Affairs Medical Center, Minneapolis, Minnesota.

Acknowledgment: The authors thank Indy Rutks and James Tacklind for their assistance in the literature search.

Grant Support: By the American College of Physicians.

Potential Conflicts of Interest: Disclosures can be viewed at www.acponline.org/authors/icmje/ConflictOfInterestForms.do?msNum=M10-1375.

Requests for Single Reprints: Frank A. Lederle, MD, Center for Chronic Disease Outcomes Research, Minneapolis Veterans Affairs Medical Center (111-O), One Veterans Drive, Minneapolis, MN 55417; e-mail, frank.lederle@va.gov.

Current author addresses and author contributions are available at www.annals.org.

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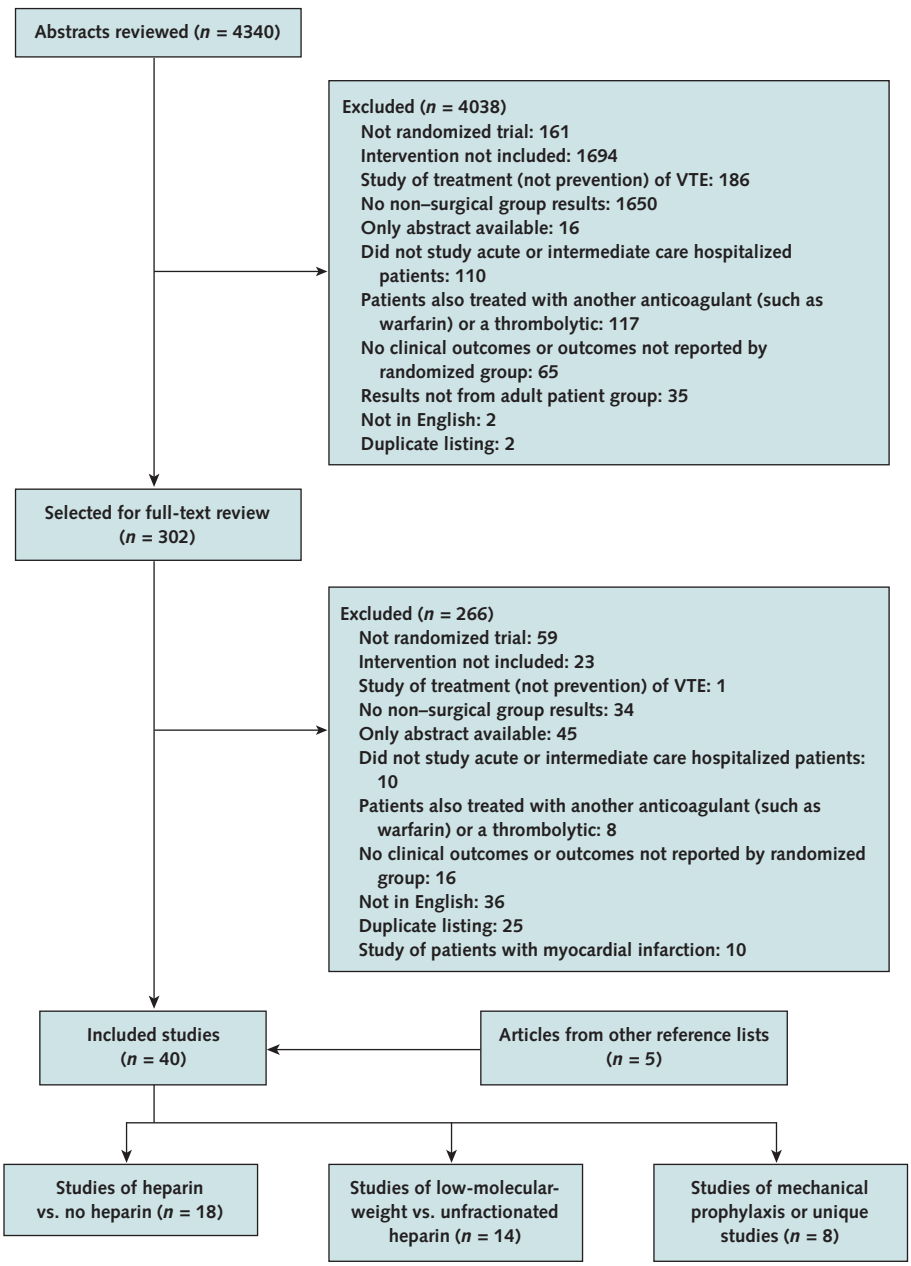
Current Author Addresses: Drs. Lederle and Wilt and Mr. MacDonald: Center for Chronic Disease Outcomes Research, Minneapolis Veterans Affairs Medical Center (111-O), One Veterans Drive, Minneapolis, MN 55417.
Dr. Zylla: Hematology Section, Minneapolis Veterans Affairs Medical Center (111-O), One Veterans Drive, Minneapolis, MN 55417.

Author Contributions: Conception and design: F.A. Lederle, T.J. Wilt.
Analysis and interpretation of the data: F.A. Lederle, D. Zylla, R. MacDonald, T.J. Wilt.
Drafting of the article: F.A. Lederle, R. MacDonald, D. Zylla.
Critical revision of the article for important intellectual content: F.A. Lederle, T.J. Wilt.
Final approval of the article: F.A. Lederle, D. Zylla, R. MacDonald, T.J. Wilt.
Statistical expertise: R. MacDonald, T.J. Wilt.
Obtaining of funding: T.J. Wilt.
Administrative, technical, or logistic support: T.J. Wilt.
Collection and assembly of data: F.A. Lederle, D. Zylla, T.J. Wilt.

Appendix Figure 1. **Summary of evidence search and selection.**



VTE = venous thromboembolism.

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Appendix Table 1. Overview of Included Studies

Study, Year (Reference)	Region	Funding Source	Study Characteristics							Study Quality			
			Indication for PA	Patient Characteristics	Study Criteria	Sample Size, *n*	Interventions	Duration per Protocol	Study Period	Allocation Concealment	Blinding	Intention-to-Treat Analysis	Withdrawals and Dropouts Adequately Described
Heparin vs. no heparin													
Medical patients													
Weber et al, 2008 (23)	Switzerland	None stated	Cancer	Mean age, 70 y (range, 55–88 y) Men, 50%	Inclusion: admitted to center of continuous care with an estimated life expectancy ≥6 mo Exclusion: VTE within 6 mo, active bleeding, creatinine clearance <20 mL/min per 1.73 m^2, thrombocytopenia, history of heparin thrombocytopenia, PTT >45 s, or PT <35% and concomitant anticoagulation on admission	20	Nadroparin, 2850 U/d (weight <70 kg) or 3800 U/d (weight >70 kg), or usual care	Not reported	90 d (treatment + follow-up)	Unclear (sealed envelopes were opened by the principal investigator to allow treatment allocation)	None	Yes	NR
Cohen et al, 2006 (22)	Multinational	Industry	CHF (NYHA class III or IV) or acute respiratory, inflammatory, or infectious disease	Mean age, 75 y (range, 53–96 y) Men, 42%	Inclusion: indications as listed, aged ≥60 y, and expected to remain in bed for ≥4 d Exclusion: high risk for bleeding; acute bacterial endocarditis; cerebral metastasis; recent hemorrhagic or ischemic stroke; brain, spinal, or ophthalmologic surgery; indwelling intrathecal or epidural catheter; serum creatinine level >180 μmol/L (>2.04 mg/dL) in a well-hydrated patient; documented hypersensitivity to contrast media; anticipated intubation for >24 h; use of antithrombotics ≤48 h before random assignment; indication for anticoagulant prophylaxis or therapy; or life expectancy <1 mo	849	Fondaparinux, 2.5 mg/d, or placebo	6–14 d (median, 7 d)	32 d	Adequate	Double	No	No
Lederle et al, 2006 (10)	United States	Nonindustry	Hospitalization in general medical unit	Mean age, 72 y Men, 99% White, 79%	Inclusion: admitted or transferred to medical service of VA medical center on day of random assignment or the previous day; aged ≥60 y; and remaining under care of VA medical service ≥3 d from random assignment Exclusion: receiving or requiring anticoagulation for reasons other than VTE prophylaxis; known thrombocytopenia; hypertension; other contraindication to low-dose heparin, in the opinion of the patient's physicians; "supportive or palliative care only" status; or occurrence of myocardial infarction, stroke, major surgery (defined as requiring general, spinal, or epidural anesthesia and lasting >30 min), or any eye surgery within the past 30 d	280	Enoxaparin, 40 mg/d, or placebo	Until discharge (mean, 12 d)	90 d	Adequate	Double	Yes	Yes
Mahé et al, 2005 (13)	Multinational	Industry	CHF or acute or respiratory disease	Mean age, 76 y Men, 41%	Inclusion: age ≥40 y, hospitalized <24 h because of acute medical illness, and immobilization Exclusion: conditions that could increase the risk for hemorrhage (hypertension, active gastroduodenal ulcer, renal failure, PT <50%, or platelet count $<50 \times 10^9$ cells/L), conditions that required full-dose anticoagulation, stroke or major surgery ≤30 d and anticoagulant or antiplatelet therapy ≤7 d, or pregnancy	2474	Nadroparin, 7500 U/d, or placebo	21 d or until discharge (mean, 13 d)	21 d	Adequate	Double	Yes	NR

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Appendix Table 1—Continued

Study, Year (Reference)	Region	Funding Source	Study Characteristics							Study Quality			
			Indication for PA	Patient Characteristics	Study Criteria	Sample Size, *n*	Interventions	Duration per Protocol	Study Period	Allocation Concealment	Blinding	Intention-to-Treat Analysis	Withdrawals and Dropouts Adequately Described
Leizorovicz et al, 2004 (7), and Kucher et al, 2005 (55)	Multinational	Industry	Acute CHF, acute respiratory failure, infectious disease, acute rheumatologic disorders, or inflammatory bowel disease	Mean age, 69 y Men, 48%	Inclusion: age ≥40 y, acute medical condition that required hospitalization ≥4 d, and ≤3 d of previous immobilization Exclusion: acute coronary syndrome within the previous month, a major surgical or invasive procedure in the previous month or to be done within the next 2 wk, bacterial endocarditis, immobilized lower limb because of a cast or fracture, stroke ≤3 mo, high risk for bleeding, platelet count <100 × 10^9 cells/L, heparin or LMWH prophylaxis >48 h before random assignment, contraindication to heparin anticoagulation, creatinine level >176.8 μmol/L (>2.0 mg/dL), hepatic insufficiency or active hepatitis, pregnancy or breastfeeding, or life expectancy <1 mo	3706	Dalteparin, 5000 U/d, or placebo	14 d (median, 14 d)	90 d	Not defined	Double	No	No
Fraisse et al, 2000 (6)	France	Industry	Acute decompensated COPD on mechanical ventilation	Mean age, 68 y (PA group, 69 y; control group, 67 y; $P = 0.020$) Men, 78%	Inclusion: age 40–80 y and weight 45–110 kg Exclusion: confirmed DVT within 6 mo or signs of DVT on Doppler ultrasonography at inclusion; an organic lesion that could bleed (active gastroduodenal ulcer or recent hemorrhagic CVA); severe liver failure leading to a decrease of PT to <50%; severe renal impairment; confirmed or uncontrolled hypertension; congenital or acquired coagulation disorder; history of hypersensitivity or thrombocytopenia to heparins of any type; contraindication to anticoagulation, venography, or angiography; or receiving acetylsalicylic acid, ticlopidine, or oral anticoagulants	223	Nadroparin, 3800–5700 U/d, or placebo	21 d or until weaned from mechanical ventilation (mean, 11 d)	11 d	Adequate	Double	No	Yes
Samama et al, 1999 (5), and Alikhan et al, 2003 (54)	Multinational	Industry	CHF (NYHA class III or IV), acute or chronic respiratory disease, infectious disease, or acute rheumatologic disorders	Mean age, 73 y Men, 50%	Inclusion: age ≥40 y, hospitalized ≥6 d, and not immobilized ≤3 d Exclusion: stroke or major surgery within 3 mo; contraindications to the use of iodinated contrast medium; known thrombophilia; creatinine level >150.20 μmol/L (>1.7 mg/dL); HIV infection; uncontrolled arterial hypertension, active peptic ulcer, bacterial endocarditis, or other conditions that could increase risk for hemorrhage; hypersensitivity to heparin or heparin-induced thrombocytopenia; platelet count <100 × 10^9 cells/L, prolonged aPTT, PT <50%, or an international normalized ratio >1.2; required anticoagulation or received any type of anticoagulation for >48 h; and pregnancy or women of childbearing years	1102	Enoxaparin, 20 mg/d; enoxaparin, 40 mg/d; or placebo	6–14 d (median, 7 d)	110 d	Adequate	Double	No	Yes
Gärdlund, 1996 (8)	Sweden	Mainly nonindustry	Infectious disease	Mean age, 76 y Percentage of men not reported	Inclusion: age ≥55 y Exclusion: ongoing anticoagulation, readmission 60 d from random assignment, active bleeding, coagulation disorder, dialysis, liver failure, HIV infection, or terminal disease	11 693	UFH, 5000 U twice daily, or usual care	21 d or until discharge (mean, 7 d)	60 d	Not defined	None	Yes	Yes

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Study, Year (Reference)	Region	Funding Source	Study Characteristics							Study Quality			
			Indication for PA	Patient Characteristics	Study Criteria	Sample Size, *n*	Interventions	Duration per Protocol	Study Period	Allocation Concealment	Blinding	Intention-to-Treat Analysis	Withdrawals and Dropouts Adequately Described
Dahan et al, 1986 (4)	France	None stated	Heart failure, respiratory diseases, malignant disease, infectious disease, or other	Mean age, 80 y Men, 62%	Inclusion: age ≥65 y and nonsurgical inpatient Exclusion: ongoing anticoagulant or platelet inhibitor therapy; active bleeding, including cerebral hemorrhage; coagulation disorders; short-term hospitalization (<7 d); thyroid diseases; or iodine allergy	270	Enoxaparin, 60 mg/d, or placebo	10 d or until discharge	10 d	Not defined	Double	No	Yes
Belch et al, 1981 (21)	Scotland	None stated	Heart failure or chest infection	Mean age, 66 y Men, 69%	Inclusion: age ≥40–80 y Exclusion: definite risk for bleeding, DVT or PE on admission, iodine allergy, or confined to bed >2 d before admission	100	UFH, 5000 U 3 times daily, or usual care	14 d or until discharge (mean, 9 d)	14 d	Not defined	DVT diagnosed by person unaware of treatment assignment	Yes	NR
Patients with stroke													
International Stroke Trial Collaborative Group, 1997 (42)	International	Multiple sources, mainly nonindustry	–	Age ranges: <50 y, 5%; 50–59 y, 11%; 60–69 y, 23%; 70–79 y, 35%; >80 y, 26% Men, 54%	Inclusion: evidence of acute stroke within 48 h, no evidence of ICH, and no clear indications for or contraindications to aspirin or heparin Exclusion: ongoing anticoagulation, small likelihood of worthwhile benefit (symptoms likely to resolve in a few hours or patient severely disabled before the stroke), or high risk for adverse effects (hypersensitivity to aspirin, active peptic ulcer, or recent GI bleeding)	14 578	UFH, 5000 U twice daily (a higher-dose group was not included in this review), or "avoid heparin"	14 d or until discharge (mean, 11 d)	6 mo; outcomes data for low-dose heparin for the first 14 d	Adequate	Open-label	Partially (>99%)	Yes
Kay et al, 1995 (41)	Hong Kong	Nonindustry and industry	–	Mean age, 67 y Men, 58% Asian (Chinese), 100%	Inclusion: diagnosis of acute stroke within the previous 48 h and aged <80 y Exclusion: CT evidence of ICH, transient neurologic deficits, sustained hypertension, major confounding neurologic or systemic illness (including a previous disabling stroke), recent major operation or known tendency toward bleeding, current anticoagulation or valvular heart disease necessitating such therapy, known hypersensitivity or any other adverse reaction to heparin, stroke but no motor deficit, or death considered to be imminent	312	Nadroparin, 4100 U/d (low-dose; a higher-dose group was not included in this review) or placebo	10 d	90 d	Adequate	Double	No	Yes
Sandset et al, 1990 (40)	Norway	Nonindustry and industry	–	Mean age, 75 y Men, 45%	Inclusion: diagnosis of acute stroke within 72 h Exclusion: comatose, hemorrhagic stroke on CT scan, stroke onset >72 h before inclusion, strokes qualifying for heparin therapy (mostly progressive or of embolic origin), bleeding diathesis, severe hypertension, severe renal failure, severe liver failure, severe anemia, thrombocytopenia, or cancer	103	Dalteparin, 3000–5500 U/d (based on body weight), or placebo	14 d or until discharge	14 d (mortality, 28 d)	Adequate	Double	No	Yes
Prins et al, 1989 (39)	The Netherlands	None stated	–	Median age range, 71–80 y Men, 52%	Inclusion: ischemic stroke within 72 h Exclusion: ongoing anticoagulation or comatose	60	Dalteparin, 2500 U twice daily, or placebo	14 d	28 d	Unclear (numbered syringes)	Double	Yes for most outcomes	NR
Dickmann et al, 1988 (38)	Germany	None stated	–	Mean age, 61 y Men, 50%	Inclusion: diagnosis of acute stroke within previous 24 h Exclusion: bleeding diathesis, hypertension, or deep coma with signs of brain herniation	46	UFH, 5000 U 3 times daily, or no heparin (usual care; these patients received heparin at day 10 after CVA)	6 d (at day 4)	10 d	Not defined	None	Yes	NR

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Turpie et al, 1987 (37)	Canada	Nonindustry and industry	–	Mean age, 69 y (range, 28–90 y) Men, 53%	Inclusion: diagnosis of acute stroke Exclusion: ongoing anticoagulation; CT evidence of hemorrhagic stroke; nonparalytic stroke; assessment of qualifying stroke >7 d after onset; stroke thought to be embolic in origin, thus requiring anticoagulation; acute DVT; history of subarachnoid hemorrhage; bleeding disorder; sensitivity to iodine or contrast dye; severe liver or renal dysfunction; or GI bleeding or active peptic ulcer	75	Danaparoid, 1000 U via IV load, then 750 U twice daily; or placebo	14 d or until discharge (mean, 12 d)	90 d	Not defined	Double	Partially	Yes
McCarthy and Turner, 1986 (36)	United Kingdom	Nonindustry and industry	–	Mean age, 76 y Men, 43%	Inclusion: diagnosis of stroke within previous 48 h Exclusion: bleeding diathesis, hypertension, grade 3 or 4 hypertensive retinopathy, history of subarachnoid hemorrhage, active peptic ulcer, allergy to iodine, goiter or thyrotoxicosis, recent myocardial infarction, or cancer	305	UFH, 5000 U 3 times daily, or no heparin (usual care)	14 d	84 d	Not defined	None	Yes	NR
McCarthy et al, 1977 (35)	United Kingdom	None stated	–	Mean age, 79 y Men, 34%	Inclusion: diagnosis of stroke within previous 48 h Exclusion: blood in cerebrospinal fluid, bleeding diathesis, hypertension, grade 3 or 4 hypertensive retinopathy, history of subarachnoid hemorrhage, history of active peptic ulcer, allergy to iodine, goiter or thyrotoxicosis, or recent myocardial infarction	32	UFH, 5000 U 3 times daily, or no heparin (usual care)	14 d	28 d	Not defined	None	Yes	NR
LMWH vs. UFH Medical patients													
Cook et al, 2011 (30)	United States, Canada, Australia, Brazil, Saudi Arabia, and the United Kingdom	Nonindustry	Critically ill patients (mainly respiratory, followed by sepsis and GI or cardiovascular conditions)	Mean age, 61 y Men, 57%	Inclusion: age ≥18 y, weight ≥45 kg, and expected to remain in ICU for ≥3 d Exclusion: major trauma, neurosurgery or orthopedic surgery, need for therapeutic anticoagulation, heparin administration in the ICU for ≥3 d, contraindication to heparin or blood products, pregnancy, life-support limitation, or enrollment in a related trial	3764	Dalteparin, 5000 U/d, or UFH, 5000 U twice daily	Median duration of ICU stay, 9 d (range, 6–16 d); median duration of exposure to study drug, 7 d (range, 4–12 d)	Median time in hospital, 21 d (range, 13–41 d)	Adequate	Double, outcome assessor	No (those who withdrew consent were excluded)	Yes
Riess et al, 2010 (32)	Germany	Industry	Acute nonsurgical disease	Mean age, 79 y Men, 41%	Inclusion: acute medical condition, aged ≥70 y, and expected significant decrease in mobility ≥4 d Exclusion: immobilization ≥3 d before random assignment, immobilization due to cast or fracture, expected major surgical or invasive procedure within 3 wk after randomization, severe sepsis or need for ventilatory support, LMWH or heparin administration for >48 h in the 5 d before random assignment, indication for anticoagulation or thrombolysis, life expectancy <6 mo or illness with very high acute mortality (>30%), acute symptomatic DVT or PE, acute or history of HIT type II, acute or history of nonhemorrhagic stroke (<3 mo), hemorrhagic stroke or ICH (<12 mo), acute or ongoing intracranial disease, high risk for GI bleeding, spinal or epidural anesthesia, lumbar puncture within the past 12 h, uncontrolled hypertension, severe liver or renal disease, acute endocarditis, known active retinopathy, and intravitreal or other intraocular bleeding	3239	Certoparin, 3000 U/d, or UFH, 5000 U 3 times daily	9.1 ± 3.4 d	90 d	Adequate	Double, outcome assessor	No (≥1 dose)	Yes

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Schellong et al, 2010 (31)	Germany	Industry	Acute nonsurgical disease	Mean age, 71 y Men, 48%	Inclusion: age ≥40 y, hospitalized for acute nonsurgical disease, and significant recent decrease in mobility (completely bedridden or only able to walk short distances with the support of nurse) Exclusion: indication for anticoagulant or thrombolytic therapy; international normalized ratio >1.5 or PTT >1.5 times the upper limit of normal; major surgical or invasive procedure within 4 wk before random assignment or expected major surgical or invasive procedure within 2 weeks of random assignment; immobilization due to cast or fracture of the lower extremity or >3 d before random assignment; heparin administration for >36 h before random assignment; acute ischemic stroke; hemorrhagic stroke or other ICH (acute or within past 12 mo); history of or current intracranial disease; injury to or operation on central nervous system, eye, or ear; uncontrolled hypertension; life expectancy <1 y; active peptic ulcer; active major bleeding within past 4 wk; hemorrhagic diathesis; lack of coagulation factors; pancreatic diseases or suspicion of malignant tumor with enhanced risk for bleeding; history of or current HIT type II; platelet count <100 × 10^9 cells/L; severe renal or hepatic disease; renal or ureteral calculus; retinopathy; intravitreal or other intraocular hemorrhage; of childbearing age (unless postmenopausal or using highly effective form of birth control); pregnancy or lactation; recent history of addictive disorder; or use of other investigational drugs at time of enrollment	342	Certoparin, 3000 U/d, or UFH, 7500 units twice daily	10 ± 2 d (mean, 8.5 d)	90 d	Adequate	Outcome assessor (ultrasonography readings)	No (≥1 dose)	Yes
Kleber et al, 2003 (29)	Germany	Industry	CHF (NYHA class III or IV) or severe respiratory disease	Mean age, 70 y Men, 52%	Inclusion: age ≥18 y, hospitalized for severe respiratory disease or heart failure, and confined to bed for more than two thirds of each day Exclusion: advanced AIDS; contraindications for LMWH or UFH therapy; hypersensitivity to contrast media; severe pancreatic, hepatic, or renal disease; severe arterial hypertension; ICH or hemorrhagic stroke within 6 mo; ocular or central nervous system surgery within 4 wk; coagulation disorders; drug or alcohol abuse; acute signs of DVT or PE; GI ulcer; or immobilized for >24 h before enrollment	668	Enoxaparin, 40 mg/d, or UFH, 5000 U 3 times daily	10 ± 2 d (mean, 10 d)	10 d	Not defined	Outcome assessor	No	Yes

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Bergmann and Neuhart, 1996 (27)	France	None stated; 1 author and 1 statistician were affiliated with industry	Cardiac or respiratory failure, cancer, infectious disease, or other	Mean age, 83 y Men, 28%	Inclusion: age ≥65 y, acute medical illness leading to decrease in autonomous mobility (<4 d), and unable to walk 10 m without assistance Exclusion: use of heparin, oral anticoagulants, aspirin, or NSAIDs >24 h before enrollment; contraindication to anticoagulation (uncontrolled hypertension, recent hemorrhagic stroke, or any other hemorrhagic disease); ongoing venous, arterial, or cardiac disease requiring anticoagulation; history of allergy or thrombocytopenia induced by UFH or LMWH; hemostasis abnormality; renal failure; contraindication to isotopic or venographic investigations; or local disorders of lower limbs likely to interfere with fibrinogen uptake test	442	Enoxaparin, 20 mg twice daily, or UFH, 5000 U twice daily	10 d (mean, 10 d)	10 d	Adequate	Double	No (≥1 dose)	Yes
Harenberg et al, 1996 (26)	Germany	None stated	Cancer or cardiovascular, pulmonary, neurologic, or other diseases	Mean age, 70.5 y Men, 45%	Inclusion: age 50–80 y, increased risk for thromboembolism, and expected bed rest for 10 d Exclusion: intolerance to heparin, acute DVT, thrombocytopenia, hereditary or acquired coagulation disorder, pretreatment with heparin other than study medication, regular intake of medication influencing blood coagulation, disease with unfavorable short-term diagnosis, septicemia, disseminated intravascular coagulation, fixed hypertension, history of any bleeding; creatinine level >265.2 μmol/L (>3 mg/dL); or PT <60%	1968	Nadroparin, 36 mg/d, or UFH, 5000 U 3 times daily	10 d	10 d	Not defined	Double	No	Yes
Lechler et al, 1996 (28)	Germany and Austria	None stated	Cancer or cardiovascular, pulmonary, endocrinologic, neurologic, or other diseases	Mean age, 74 y Men, 38%	Inclusion: age ≥18 y, increased risk for thromboembolism, and expected immobilization for more than one half of the daytime for 7-d study period Exclusion: thromboembolism on admission; anticoagulation, aggregation inhibitors, or NSAIDs within 7 d; regional anesthesia; pregnancy or lactation; bleeding disorder; thrombocytopenia; head trauma within 6 mo; recent hemorrhagic stroke; endocarditis; suspicion of internal bleeding; severe liver disease; or creatinine level >265.2 μmol/L (>3 mg/dL)	959	Enoxaparin, 40 mg twice daily, or UFH, 5000 U 3 times daily	7 d	7 d	Not defined	Double	Yes	No
Aquino et al, 1990 (25)	France	None stated	Various, including CHF, venous insufficiency, cancer, and obesity	Mean age, 84 y Men, 15%	Inclusion: age >70 y and increased risk for thromboembolism Exclusion: anticoagulation abnormalities, presence of DVT, or contraindications to prophylactic heparin therapy	99	Nadroparin, 7500 U/d, or UFH, 5000 U (weight <60 kg) or 7500 U (weight >60 kg) twice daily	Not reported	10 d	Not defined	None, open-label	Yes	NR
Harenberg et al, 1990 (24)	Germany	None stated	Various, including cancer, heart insufficiency, coronary heart disease, atrial fibrillation, infectious disease, or other	Mean age, 66 y Men, 45%	Inclusion: age 40–80 y, increased risk for thromboembolism, and expected bed rest for ≥7 d Exclusion: cerebral hemorrhage within 6 mo, underlying bleeding disorder, hypertension, cirrhosis of the liver, renal insufficiency, increased risk for GI or urogenital bleeding, acute pancreatitis, disseminated intravascular coagulation, known intolerance to heparin, or indications for anticoagulation or fibrinolytic or antiplatelet therapy	200 (final size of treatment groups unclear)	Dalteparin, 1.5 aPTT U/d, or UFH, 5000 U 3 times daily	10 d (range, 7–12 d)	10 d	Not defined	Double	No (patients treated <7 d were excluded from analyses)	Yes

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Patients with stroke Sherman et al, 2007 (47)	United States	Industry	–	Mean age, 66 y Men, 56% White, 59%; black, 7%; Asian, 21%; Hispanic, 8%; other 4%	Inclusion: age ≥18 y, acute ischemic stroke, and NIHSS score ≥2 for motor function of the leg Onset of stroke symptoms had to have occurred within 48 h before random assignment Exclusion: evidence of VTE at screening or evidence of active bleeding; evidence or history of ICH; heparin- or enoxaparin-induced thrombocytopenia or thrombosis, or both; hypersensitivity to iodinated contrast media or iodine; spinal or epidural analgesia or lumbar puncture within 24 h; thrombolytic treatment within the preceding 24 h; comatose at screening (NIHSS score ≥2 for level of consciousness); known or suspected cerebral aneurysm or arteriovenous malformation; confirmed cancer that might have posed an increased risk for bleeding or compromise follow-up or outcome assessment; impaired hemostasis; major surgery or major trauma within 3 mo; expected need for full-dose treatment with therapeutic levels of an anticoagulant; treatment with LMWH or UFH at a prophylactic dose >48 h before inclusion; allergy or known hypersensitivity to heparin or enoxaparin; bacterial endocarditis; prosthetic heart valve; known or suspected severe anemia; uncontrolled arterial hypertension or clinical hypertensive urgency; life expectancy <3 mo because of comorbid disorders; clinically relevant serious diseases, including severe liver disease or renal failure; lactation, pregnancy, or of childbearing age	1762	Enoxaparin, 40 mg/d, or UFH, 5000 U twice daily	10 d (range, 6–14 d; mean, 10.5 d [SD, 3.2])	90 d	Adequate	Open-label, blinded adjudication of major hemorrhage events	No	Yes
Diener et al, 2006 (46)	Europe	Industry	–	Mean age, 67 y Men, 57%	Inclusion: age 18–85 y and acute ischemic stroke within 24 h of random assignment Severity defined by NIHSS score of 4 to 30 with mild to severe paresis of the leg Exclusion: indication for thrombolysis; no availability of CT; CT-documented signs of intracerebral or subarachnoidal hemorrhage; current bleeding or thrombosis; history of bleeding or thrombosis within 12 mo; recurrent GI ulcerations; postthrombotic syndrome; acute or unstable cardiovascular disease; major infection; active, recurrent, or metastatic cancer within 5 y; platelet count $<75 \times 10^9$ cells/L at baseline; known hypersensitivity to heparin; known severe diabetic retinopathy; estimated body weight <55 kg; or pregnancy or lactation	545	Certoparin, 3000 U/d, or UFH, 5000 U 3 times daily	12–16 d (mean, 13.6 d [SD, 3.2])	90 d	Not defined	Double	Yes	Yes

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Hillbom et al, 2002 (45)	Finland	Industry	–	Mean age, 69 y Men, 60%	Inclusion: age 18–90 y; acute ischemic stroke within 48 h of random assignment, with lower-limb paralysis lasting ≥24 h; and consciousness Exclusion: unconscious (Glasgow Coma Scale score <9) or immobilized before the onset of stroke; evidence of either hemorrhagic stroke or stroke thought to be cardioembolic in origin; evidence or history of DVT, PE, or myocardial infarction or recent neurosurgery (≤3 mo); history of subarachnoid hemorrhage; GI bleeding or active peptic ulceration; hypersensitivity to heparin, LMWH, or radio-opaque contrast media; severe heart failure; uncontrolled arterial hypertension; hepatic or renal impairment; cancer; endocarditis; hemorrhagic diathesis; current drug abuse; any clinical condition requiring anticoagulant or antiplatelet therapy; lactation or pregnancy; abnormal blood clotting test results; other investigational medication within 3 mo; or study treatment could not be started within 48 h of the onset of the qualifying stroke. Concomitant treatment with anticoagulant or antithrombotic therapy, NSAIDs, aspirin, other antiplatelet therapy, or any other treatment that may have influenced the interpretation of the study data was prohibited	212	Enoxaparin, 40 mg/d, or UFH, 5000 U 3 times daily	10 ± 2 d or until discharge	90 d	Not defined	Double	No	Yes
Dumas et al, 1994 (43)	Europe	None stated	–	Mean age, 73 y Men, 42%	Inclusion: acute ischemic stroke (within 72 h of random assignment) with leg paralysis Exclusion: hemorrhagic stroke, ischemic stroke if screening was not possible within 72 h after onset, recent GI bleeding or active or duodenal ulcer, allergy to iodine or heparin, DVT present at screening, bleeding tendency, history of subarachnoid hemorrhage, moderate or severe renal insufficiency, hypertensive retinopathy, or hypertension	179	Danaparoid, 1250 U/d, or UFH, 5000 units twice daily	9 d (range, 9–13 d)	90 d	Not defined	Double, blinded adjudication of hemorrhage events	No	Yes
Turpie et al, 1992 (44)	Canada	Nonindustry and industry	–	Mean age, 72 y (range, 36–94 y) Men, 44% White, 97%; other, 3%	Inclusion: acute ischemic stroke resulting in lower-limb paralysis Exclusion: hemorrhagic stroke on CT; stroke of embolic origin, thus requiring anticoagulation; history of subarachnoid hemorrhage; established acute DVT; assessed >7 d after onset of qualifying stroke and had an abnormal impedance plethysmography; bleeding disorder; severe peptic ulcer within 4 wk; or sensitive to iodine or contrast dye	87	Danaparoid, 750 U twice daily, or UFH, 5000 units twice daily	14 d or until discharge (mean, 12 d [range, 1–15 d])	90 d	Not defined	Double, blinded adjudication of major hemorrhage events	Partially (during treatment period)	Yes
Mechanical vs. no mechanical prophylaxis													
Dennis et al, 2009 (50)	United Kingdom, Italy, Australia	Nonindustry	Acute stroke	Mean age, 76 y (range, 68–83 y) Men, 49%	Inclusion: patients admitted to hospital within 1 wk of an acute stroke and immobile (unable to walk independently to the toilet) Exclusion: stroke due to subarachnoid hemorrhage, peripheral vascular disease, diabetic or sensory neuropathy, or responsible clinician or nurse judged that graduated compression stockings might cause skin damage	2518	Thigh-high graduated compression stockings or no ("avoid") graduated compression stockings	30 d or until discharge, mobility, or later second compression Doppler ultrasonography	30 d	Adequate	Open, investigators were blinded to primary outcomes	Yes	Yes

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Muir et al, 2000 (49)	United Kingdom	Nonindustry	Acute stroke	Mean age, 76 y Percentage of men not reported	Inclusion: acute ischemic stroke within 72 h of random assignment, not independently ambulatory ≤24 h of admission, and leg weakness with NIHSS score ≥1 Exclusion: comatose, life-threatening intercurrent illness, critical lower-limb ischemia, or severe dermatologic conditions	98	Full-length graded compression stockings or routine care	Not reported	9 d	Not clearly defined (sealed envelopes)	Open, blinded data review	No	Yes
Prasad et al, 1982 (48)	United Kingdom	None stated	Acute stroke	Mean age, 79 y Men, 46%	Inclusion: acute stroke within 72 h of random assignment and weakness of up to MRC grade 2/6 motor power Exclusion: comatose or other clinically unacceptable conditions	26	Pneumatic calf compression or no calf compression (control)	10 d	10 d	Not defined	Open	Yes	NR
Miscellaneous trials													
CLOTS Trial Collaboration, 2010 (53)	United Kingdom, Italy, Ireland, Australia, Czech Republic, Portugal, India, Canada, Mexico	Nonindustry	Acute stroke	Mean age, 75 y (range, 67–83 y) Men, 49%	Inclusion: hospitalized within 1 wk of an acute ischemic or hemorrhagic stroke (could be enrolled between the day of admission [day 0] and day 3 in the hospital) and immobile (could not walk independently to the toilet) Exclusion: subarachnoid hemorrhage and those with severe peripheral vascular disease or diabetic or sensory neuropathy	2518	Thigh-high or below-knee graduated compression stockings	Until discharge (although patient was removed if independently mobile or declined to wear the stockings or if staff became concerned about the patient's skin)	30 d	Adequate	Open, outcome assessors (ultrasonographers) were blinded	Yes	Yes
Hull et al, 2010 (34)	Multinational	Industry	Acutely ill elderly patients with acute infection without septic shock, acute respiratory insufficiency, heart failure, ischemic stroke, or other	Mean age, 68 y Men, 49% White, 75%; black, 7%; Hispanic, 13%; Asian, 3%; other, 2%	Inclusion: acute medical illness; aged ≥40 y; life expectancy ≥6 mo; and recently reduced mobility for ≤3 d, considered by the enrolling investigator as likely to have reduced mobility for ≥3 d after enrollment "Reduced mobility, meaning" requiring total bed rest or being sedentary without bathroom privileges (level 1 immobility) or with bathroom privileges (level 2 immobility)	6085	Enoxaparin sodium, 40 mg/d, or placebo; all patients received open-label enoxaparin, 40 mg/d, for 10 ± 4 d	28 ± 4 d	180 d	Adequate	Double	No (had to receive ≥1 dose of study treatment)	Yes
Luba et al, 2007 (33)	Poland	None stated	Acutely ill elderly patients with heart failure, severe respiratory disease, ischemic stroke, or other	Mean age range, 41–?>80 y Men, 52%	Inclusion: age >40 y, hospitalization ≥6 d, immobilization period 3–14 d, and absence of clinical or ultrasonographic symptoms of lower-extremity DVT Exclusion: ongoing anticoagulation (except aspirin), immobilizing disease within 6 mo, contraindication to LMWH (high risk for hemorrhage or allergy), cancer, mental disorders, or alcoholism	300	Nadroparin, 3800–5700 U/d (based on weight), during immobilization or during immobilization plus an additional 10 d	≤37 d (mean, 5.1 d [group 1] and 4.5 d [group 2])	90 d	Not defined	Open-label	Yes	Yes

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Lacut et al, 2005 (52)	France	Nonindustry and industry	Acute stroke	Mean age, 63 y Men, 58%	Inclusion: age >18 y and traumatic or spontaneous intracerebral hemorrhage with or without subarachnoidal hemorrhage Exclusion: extracranial or subdural hematoma, traumatic intracerebral hemorrhage due to polytrauma that included the lower limbs, hemorrhagic transformation of ischemic infarct and vasculitis (when the diagnosis was established), DVT within 3 mo, lower-limb arteriopathy with an ankle-to-arm systolic pressure index <0.70 mm Hg, venous graft, wound in lower limb related to vascular disease (ulcer or trauma), "do not resuscitate" order, or >24-h delay since hospital admission	151	Graded elastic stockings or graded elastic stockings plus intermittent pneumatic compression	10 d	90 d	Adequate	Outcome assessors	No	Yes
Green et al, 1998 (51)	United States	Nonindustry and industry	Acute stroke (~75% had received thromboembolism prophylaxis before random assignment)	Mean age, 62 y (range, 24–87 y) Men, 60%	Inclusion: patients with stroke admitted to rehabilitation center ≥7 d from the acute event and unable to walk 15 ft Exclusion: receiving warfarin, uncontrolled hypertension, platelet count $<100 \times 10^9$ cells/L or coagulopathy, severe liver disease, renal failure, history of thromboembolism, or allergy to radiographic contrast media	90	Calf compression leggings or ardeparin LMWH, 50 U/kg	≥1 wk (until discharge)	24 d (average length of stay in stroke rehabilitation)	Not defined	Open	No (had to receive therapy for ≥1 wk)	Yes

aPTT = activated partial thromboplastin time; CHF = congestive heart failure; CLOTS = Clots in Legs Or sTockings after Stroke; COPD = chronic obstructive pulmonary disease; CT = computed tomography; CVA = cerebrovascular accident; DVT = deep venous thrombosis; GI = gastrointestinal; HIT = heparin-induced thrombocytopenia; ICH = intracranial hemorrhage; ICU = intensive care unit; IV = intravenous; LMWH = low-molecular-weight heparin; MRC = Medical Research Council; NIHSS = National Institutes of Health Stroke Scale; NR = not reported; NSAID = nonsteroidal anti-inflammatory drug; NYHA = New York Heart Association; PA = prophylactic anticoagulation; PE = pulmonary embolism; PT = prothrombin time; PTT = partial thromboplastin time; UFH = unfractionated heparin; VA = Veterans Affairs; VTE = venous thromboembolism.

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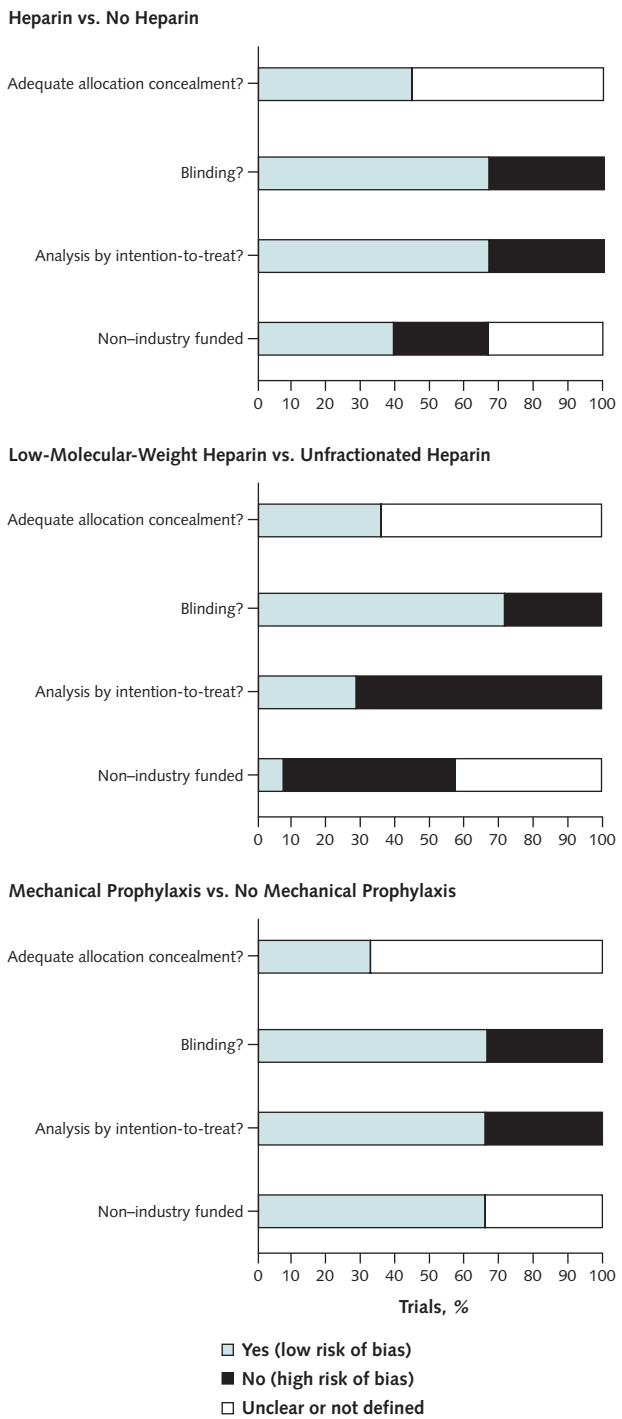
Appendix Figure 2. **Risk-for-bias graphs for trials of heparin prophylaxis versus no heparin, low-molecular-weight versus unfractionated heparin prophylaxis, and mechanical versus no mechanical prophylaxis.**



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Appendix Table 2. **Strength-of-Evidence Table for Clinical Outcomes**

Group and Outcome	Studies, *n*	Participants, *n*	Risk for Bias*	Consistency	Directness	Precision	Peto Odds Ratio (95% CI)†	Strength of Evidence
Heparin vs. no heparin								
Mortality								
Medical patients	10	20 717	Medium	Consistent	Direct	Precise	0.94 (0.84–1.04)	Moderate
Patients with stroke	8	15 405	Medium	Consistent	Direct	Imprecise	0.91 (0.70–1.18)	Low
Symptomatic DVT								
Medical patients	5	5957	Medium	Consistent	Direct	Imprecise	0.78 (0.45–1.35)	Low
Patients with stroke	1	206	Medium	Unknown	Direct	Imprecise	0.14 (0.00–7.09)	Insufficient
Pulmonary embolism								
Medical patients	10	20 717	Medium	Consistent	Direct	Precise	0.69 (0.52–0.90)	Moderate
Patients with stroke	5	14 862	Medium	Consistent	Direct	Imprecise	0.72 (0.50–1.04)	Low
All bleeding events								
Medical patients	8	8744	Medium	Consistent	Direct	Precise	1.34 (1.08–1.66)	Moderate
Patients with stroke	6	522	Medium	Consistent	Direct	Imprecise	0.95 (0.55–1.63)	Low
Major bleeding events								
Medical patients	9	20 447	Medium	Consistent	Direct	Imprecise	1.49 (0.91–2.43)	Low
Patients with stroke	8	15 405	Medium	Consistent	Direct	Precise	1.66 (1.20–2.28)	Moderate
LMWH vs. UFH								
Mortality								
Medical patients	9	11 650	Medium	Consistent	Direct	Precise	0.91 (0.73–1.13)	Moderate
Patients with stroke	5	2785	Medium	Consistent	Direct	Precise	1.00 (0.81–1.22)	Moderate
Symptomatic DVT								
Medical patients	3	3753	Medium	Unknown	Direct	Imprecise	1.00 (0.20–4.94)	Low
Patients with stroke	2	1974	Medium	Consistent	Direct	Imprecise	0.34 (0.11–1.00)	Low
Pulmonary embolism								
Medical patients	9	11 650	Medium	Consistent	Direct	Imprecise	0.70 (0.44–1.11)	Low
Patients with stroke	4	2698	Medium	Consistent	Direct	Imprecise	0.57 (0.25–1.34)	Low
All bleeding events								
Medical patients	8	11 551	Medium	Consistent	Direct	Precise	0.90 (0.76–1.05)	Moderate
Patients with stroke	5	2785	Medium	Consistent	Direct	Imprecise	0.94 (0.71–1.25)	Low
Major bleeding events								
Medical patients	9	11 650	Medium	Consistent	Direct	Imprecise	0.89 (0.70–1.15)	Low
Patients with stroke	5	2785	Medium	Consistent	Direct	Imprecise	1.49 (0.73–3.06)	Low
Mechanical vs. no mechanical prophylaxis								
Patients with stroke								
Mortality	3	2641	Medium	Consistent	Direct	Imprecise	1.13 (0.89–1.43)	Low
Skin disruptions	1	2518	Low	Unknown	Direct	Precise	4.02 (2.34–6.91)	Moderate

DVT = deep venous thrombosis; LMWH = low-molecular-weight heparin; UFH = unfractionated heparin.
* On the basis of the aggregate quality of the included trials (low risk = generally good quality; medium risk = generally fair quality; high risk = generally poor quality).
† Risk ratio was used when event rates were >1% (mortality, any bleeding events, and skin damage).

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